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Stellar populations of quasar host galaxies with MFICA decomposition

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ABSTRACT

Galaxy evolution theories require co-evolution between accreting supermassive black holes (SMBH) and galaxies to explain many properties of the local galaxy population, yet observational evidence for the mechanisms driving this co-evolution is lacking. The recent star formation histories of the host galaxies of accreting SMBHs [active galactic nuclei (AGNs)] can help constrain the processes that feed SMBHs and halt star formation in galaxies, but are difficult to obtain for the most luminous AGNs (quasars). We introduce Mean-Field Independent Component Analysis (MFICA) to decompose quasar spectra and obtain recent star formation histories of their host galaxies. Applying MFICA to quasar spectra from the Sloan Digital Sky Survey DR7 Quasar Catalogue in the redshift range $0.16 \leq z \leq 0.76$, we find that 53 per cent of quasar host galaxies are star-forming, 17 per cent lie in the green-valley, and only 5 per cent are quiescent. This contrasts with 14, 11, and 74 per cent of a mass-matched control sample that are star-forming, green-valley, and quiescent, respectively. We find that $\sim$ 25 per cent of quasars are hosted by post-starburst galaxies, an excess of 28 ± 1 compared to our control sample. While the heterogeneity of recent star formation histories implies multiple SMBH feeding mechanisms, the excess of post-starburst host galaxies demonstrates the link between accreting SMBHs and a recent starburst followed by rapid quenching. Given that massive post-starburst galaxies are predominantly caused by gas-rich major mergers, our results indicate that 30–50 per cent of quasars originate from merger-induced starbursts.

Key words: galaxies: evolution – quasars: general – galaxies: star formation – galaxies: active.

1 INTRODUCTION

A central challenge in extragalactic astronomy is characterizing the mechanisms responsible for altering the rate of star formation in galaxies (Gabor et al. 2010) and reproducing the bi-modality of galaxy colours in the local Universe (Baldry et al. 2004). Supermassive black holes (SMBHs) hosted by most massive galaxies are implicated to play a leading role. Scaling relations between SMBH and galaxy properties, such as the $M_{\rm BH}$–σ_* (Ferrarese & Merritt 2000; Gebhardt et al. 2000) or $M_{\rm BH}$–$M_{\rm bulge}$ (Magorrian et al. 1998; Häring & Rix 2004) relations, suggest that galaxies and black holes evolve in step (Silk & Rees 1998; Kormendy & Ho 2013) or that a common phenomenon regulates their growth and evolution (Jahnke & Macciò 2011).

SMBHs can accrete gas from their surroundings, forming accretion discs that radiate across the electromagnetic spectrum in episodes known as active galactic nuclei (AGNs). Numerical simulations require some fraction of energy from an AGN to couple to the gas in a galaxy – a phenomenon known as AGN feedback (Silk & Rees 1998; Fabian 2012) – to prevent the overproduction of stars in massive galaxies (Benson et al. 2003), reproduce observed bimodality (Croton et al. 2006), and modulate the growth of black holes (Di Matteo, Springel & Hernquist 2005).

However, conclusive evidence for AGN feedback is missing, and it is unclear which galaxy scale processes (e.g. mergers, bars, disc instabilities) are most important in triggering AGNs and growing black holes (see Alexander & Hickox 2012; Fabian 2012, for reviews on AGN feedback and feeding). The crucial evidence may lie in studying the star formation histories (SFHs) of AGN host galaxies, which connect changes in galaxy-scale gas supplies to black hole accretion.

Studies on the ongoing star formation properties of AGN hosts have returned mixed results. Some studies find that AGNs are hosted by star-forming galaxies on the main sequence of star formation (Silverman et al. 2009; Mullaney et al. 2012; Rosario et al. 2013a; Trump et al. 2013; Yue et al. 2018; Zhuang & Ho 2022) with blue colours (Li et al. 2021) and young stellar populations (Kauffmann et al. 2003b; Holt et al. 2007; Bessiere et al. 2017; Burtscher et al. 2021; Ren et al. 2024). Active star formation in AGN hosts might suggest that AGN feedback is absent/inefficient, or that AGN activity promotes star formation (Zinn et al. 2013). Alternatively, active star formation may manifest from the requirement of cold gas for both AGN activity and star formation or due to star formation fuelling SMBH accretion (Ciotti & Ostriker 2007; Kauffmann & Heckman 2009; Ni et al. 2023). Star-forming AGN hosts may also be the sites of AGN feedback in the future (Bär et al. 2017; Harrison & Ramos Almeida 2024). In fact, Ward et al. (2022) show that AGNs reside in star-forming galaxies even in simulations as the impact of AGN feedback is delayed.

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Contrary to the finding of star-forming hosts, many other studies suggest that AGN hosts lie in the 'green valley' with star formation rates (SFRs) below the star formation main sequence (Salim et al. 2007; Silverman et al. 2008, 2009; Shimizu et al. 2015; Ellison et al. 2016; Leslie et al. 2016), green colours (Martin et al. 2007; Schawinski et al. 2009), and older stellar populations (Matsuoka et al. 2015). The presence of AGN host galaxies in the green valley might implicate the AGN in causing the transition of galaxies from star formation to quiescence (Silverman et al. 2008; Leslie et al. 2016). However, some studies find a delay between the end of star formation and AGN activity (Davies et al. 2007; Schawinski et al. 2007; Wild, Heckman & Charlot 2010), implying that AGN activity is merely coincident and not responsible for the quiescence of a galaxy (Hopkins 2012; Lanz et al. 2022). Finally, several studies, including some of the earliest studies of AGN host galaxy properties, find AGN hosts to be quiescent (McLure et al. 1999; Nolan et al. 2001; Birchall et al. 2023; Ni et al. 2023).

A range of factors complicates the consolidation and interpretation of these results. AGNs vary over time-scales shorter than changes in star formation in galaxies (Novak, Ostriker & Ciotti 2011). This variability and the short duty cycles of AGNs, which result in AGNs turning on and off over many cycles (e.g. Shankar, Weinberg & Miralda-Escudé 2009), can make it challenging to draw physical connections between AGN activity and host galaxy properties, even when these connections are present (Aird, Coil & Georgakakis 2019). AGN host galaxies also show different properties when selected in different wavelengths (Hickox et al. 2009; Ellison et al. 2016; Ji et al. 2022). Finally, care must be taken to account for the influence of latent physical properties, like stellar mass, which can explain differences between the properties of AGN host galaxies and their inactive counterparts (Silverman et al. 2009).

Thus, a measured statistical treatment of large samples of galaxy and AGN properties is needed to investigate AGN-galaxy co-evolution. Rest-frame optical spectra of galaxies are a rich source of information, with absorption and emission lines encoding their current stellar populations and past SFHs (see Conroy 2013, for a review on SFH modelling). Where feasible, stellar population modelling of AGN host galaxy spectra has revealed significant intermediate-age (lifetimes of < 1–2 Gyr) stellar populations dominated by A- and F-type stars (Kauffmann et al. 2003b; Vanden Berk et al. 2006; Cales et al. 2013; Canalizo & Stockton 2013; Cales & Brotherton 2015; Matsuoka et al. 2015; Riffel et al. 2023), whose presence (characterized by strong Balmer absorption) along with an absence of short-lived O and B stars is a characteristic of post-starburst, or E+A, galaxies (see French 2021, for a review on post-starburst galaxies). Though rare in the local Universe (Pawlik et al. 2018), studies have shown that a large fraction of the progenitors of quiescent galaxies may have been rapidly quenched post-starburst galaxies (Yesuf et al. 2014; Wild et al. 2020; Taylor et al. 2023). Numerical simulations widely predict the co-existence of luminous AGN activity – particularly a quasar phase – alongside post-starburst-like stellar populations (Hopkins et al. 2008; Snyder et al. 2011; Lotz et al. 2021). Results on studying the coincidence of AGN activity and the post-starburst phase in observations, however, have been mixed (Goto 2006; Yan et al. 2006; Brown et al. 2009; Yesuf et al. 2014; Lanz et al. 2022; Almaini et al. 2025).

Studies of AGN host galaxy spectra have focussed primarily on low-luminosity Type-2 AGNs (e.g. Kauffmann et al. 2003b) where a dusty torus is thought to block light from the accretion disc of the AGN, allowing modelling of the host galaxy continuum. These studies have then invoked the unified model of AGNs to argue that their results are generalizable to all AGNs. Such generalizations were adopted owing to the challenge of modelling the host galaxies of more luminous Type-1 AGNs and quasars, where an unobscured accretion disc outshines the host galaxy. The host galaxy properties of Type-1 quasars, in particular their stellar populations, are thus not well constrained, and it is unclear if their properties are the same as other less-luminous Type-2 AGNs, or perhaps different (Kim, Ho & Im 2006; Trump et al. 2013; Chen et al. 2015; Zakamska et al. 2016; Suh et al. 2019; Zou et al. 2019; Zhuang & Ho 2022). Special 'decomposition' techniques are required to disentangle the AGN and host galaxy contributions embedded in the spectra of Type-1 AGNs and quasars, and subsequently enable modelling of their host stellar populations.

1.1 Decomposing AGN spectra

Several decomposition techniques have been applied to observations of a single unresolved spectrum, such as the spectra from the Sloan Digital Sky Survey (SDSS; York et al. 2000). These decomposition techniques rely on simultaneously fitting templates – either theoretical models or empirical templates – that individually describe AGN and galaxy spectra (e.g. Buchner et al. 2024). Stellar population synthesis models are often used for the host galaxy (e.g. Cales & Brotherton 2015; Matsuoka et al. 2015; Wei et al. 2018) as they enable extraction of valuable physical properties, such as galaxy stellar mass, SFR, or stellar population age, directly from the decomposition. The contribution of the accreting black hole to the AGN spectrum can be modelled using parametric components (e.g. a power-law continuum) and/or empirical templates to reconstruct individual emission line strengths and line complexes (Temple, Hewett & Banerji 2021). Studies employing decomposition with theoretical models or empirical templates, however, are often limited in applicability to spectra with high signal-to-noise ratios (Matsuoka et al. 2015), or to spectra pre-selected with significant galaxy starlight contribution (Cales & Brotherton 2015; Wei et al. 2018).

Instead of using theoretical or empirical templates, many studies have used principal component analysis (PCA) for decomposition. PCA has been used to derive orthogonal components to reconstruct quasar (Francis et al. 1992; Yip et al. 2004b) and galaxy (Connolly et al. 1995; Ronen, Aragon-Salamanca & Lahav 1999; Yip et al. 2004a; Wild et al. 2007) spectra, which are then combined in linear decomposition to decompose quasar spectra (Vanden Berk et al. 2006; Shen et al. 2008, 2015; Jalan et al. 2023; Ren et al. 2024). PCA decomposition has a major limitation: galaxy and AGN spectra are not naturally decomposed into statistically orthogonal components, and thus PCA produces components with features of different physical origins mixed together. Irrespective of the origin of the components, it is difficult to fully account for the contamination by residual levels of host galaxy light.

In this study, we introduce the use of Mean-Field Independent Component Analysis (or MFICA) as an alternative data-driven technique for decomposing quasar spectra from SDSS. The MFICA code we use in this paper was adapted from the codes of jen Sørensen, Winther & Hansen (2002) and Opper & Winther (2005), used to classify galaxies by Allen et al. (2013), and reconstruct quasar spectra by Rankine et al. (2020). In Section 2 we give a mathematical description of MFICA, describe the selection of training and testing samples for MFICA, spectrum pre-processing, and generation of templates with MFICA. In Section 3, we explore properties of the MFICA galaxy templates. In Section 4, we validate the performance of MFICA by decomposing mock spectra. In Section 5, we apply MFICA decomposition to a sample of quasar spectra from SDSS, and discuss the implications of the results in Section 6. Throughout

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this paper, we assume a Lambda cold dark matter cosmology, with $H_0 = 70\,\mathrm{km\,s^{-1}Mpc^{-1}}$, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$. If you are more interested in the science results of this paper, we recommend skipping the technical sections and going straight to Section 5.

2 MEAN FIELD INDEPENDENT COMPONENT ANALYSIS

In Sections 2.1 and 2.2, we describe MFICA and motivate its use. In Section 2.3, we describe the collation of training and testing samples of spectra for MFICA. Finally in Section 2.4 we describe our approach to generate templates capable of reconstructing the spectra of galaxies and high-luminosity quasars using MFICA.

2.1 What is MFICA?

Independent component analysis (ICA) is an extensively applied blind source separation (BSS) technique that also possesses dimensionality reduction capabilities analogous to PCA. Given some data X, BSS techniques try to estimate the original signals/sources, S, that were mixed together in unknown proportions to produce the observed data X. BSS can be mathematically formulated as finding the matrix of sources S, from the matrix of data X, such that

$$X = AS, \tag{1}$$

where A is known as the mixing matrix.

Neither S nor A are known a priori. Thus, to solve equation (1), some very general assumption needs to be made about the sources S. In PCA, the sources S are required to be orthogonal. In ICA, the sources S are instead required to be statistically independent. ICA thus finds those sources that satisfy the inversion of equation (1) and are maximally statistically independent. Maximizing independence, however, is not straightforward, and so most ICA codes maximize some proxy of independence (see Hyvärinen, karhunen & Oja 2001, for a review on ICA). ICA has seen some use in source separation problems in astrophysics (Allen et al. 2013; Waldmann et al. 2013; Querejeta et al. 2015). It has also been used in the reconstruction of spectra, where the derived sources were found to be related to the stellar populations ages of galaxies (Kaban, Nolan & Raychaudhury 2005; Lu et al. 2006).

For observational data, such as SDSS spectra, equation (1) can be reformulated as

$$X = AS + \Sigma, \tag{2}$$

where Σ is noise present in the original data X (Sørensen et al. 2002). One approach to solving for the sources and other unknown quantities is to use Bayes Theorem, and invoke prior knowledge for the sources S (Sørensen et al. 2002). A, S, and Σ can then be found by maximizing likelihood or maximizing a posteriori probability (Sørensen et al. 2002). The assumption of independence, however, is not sufficient to solve for the unknowns this way. The true posteriors of the sources are unknown a priori.

One way of circumventing this issue is to construct a 'good' approximation to the unknown true posterior of the sources S. This can be achieved through a technique like mean-field variational theory, and is the premise for Mean-Field ICA (MFICA; see Sørensen et al. 2002, for a review).

2.2 Objectives with MFICA

We want to apply (or rather train) MFICA on samples of galaxy and quasar spectra to create a set of templates (known as components), capable of reconstructing galaxy and quasar spectra. The use of MFICA for the creation of spectral templates over other data-driven methods offers several benefits:

(i) MFICA does not require rescaling of data to unit variance and zero mean, i.e. no whitening (see Stone 2004, for more on whitening and ICA). Data whitening is counter-intuitive when the data are intrinsically composed of positive constituents, like stars in galaxy spectra. Whitening also distorts the natural shape of spectra, altering the form of features identified by template generation schemes.

(ii) Like other Bayesian inference techniques, MFICA allows the incorporation of prior information on the original components S, such as a constraint of positivity. This positivity constraint was used by Allen et al. (2013) to generate galaxy templates and by Rankine et al. (2020) to generate quasar templates. The combination of positivity and un-whitened spectra results in positive components with easily inferred physical meaning.

(iii) MFICA allows pre-specification/defining fixed components. This pre-specification and the ability to include strong priors on the sources simplifies the MFICA analysis. For example, MFICA can be applied in series to different galaxy subsamples, as we already know that galaxies are composed of different stellar populations. Differences in the relative fractions of stellar populations create distinct subsamples of galaxies.

2.3 MFICA training samples

MFICA requires both samples of galaxy and quasar spectra for the creation of components capable of reconstructing those spectra. The construction of individual training samples of galaxies and quasars is described below.

2.3.1 MFICA galaxy training sample

Galaxies used for the creation of MFICA galaxy components were taken from the SDSS data release 7 (DR7; Abazajian et al. 2009) main galaxy catalogue (Strauss et al. 2002). We selected galaxies with secure spectroscopic redshifts in the range $0.16 < z < 0.38$ and mean per pixel spectral signal-to-noise ratio (S/N) > 8, calculated over a wavelength range of 3797–8400 Å. Next, using emission line properties in the MPA-JHU catalogue (Brinchmann et al. 2004; Tremonti et al. 2004), we removed objects classified as AGNs or LINERs based on their emission-line ratios (i.e. $i_{class} = 3, 4$, and 5 in Brinchmann et al. 2004) and dusty galaxies with Balmer decrements (the ratio of Hα to Hβ emission-line fluxes) > 4.8, where each emission line is detected with S/N > 5. We also removed potential AGNs with ratio of velocity dispersions of Balmer to forbidden lines > 1.6, when all emission lines are detected with S/N ratio > 5. Finally, we removed a small number of objects with unphysical absolute r-band magnitudes and less than 3825 good pixels in their spectra. The remaining galaxy spectra make up the MFICA parent galaxy sample.

Like other artificial intelligence techniques, MFICA requires a training data set that is unbiased but also large enough to quantify intrinsic features present in galaxy spectra. The construction of the ideal training sample thus requires quantifying the diversity of spectra in the parent sample. We do this by splitting the sample into different types using the PCA catalogue of Wild et al. (2007), which provides information on the recent (≤ 1 Gyr) SFHs of galaxies. Using the Wild et al. (2007) demarcation in PC1–PC2, we split the sample into quiescent, star-forming, starburst, post-starburst, and green-valley galaxies. Selecting a similar number of galaxies

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in each class ensures that we encompass most of the variation observed in galaxy spectra, and no single galaxy class dominates the component generation. We limit the number of quiescent galaxies in the parent sample to 2000 to ensure balance in the numbers of the different galaxy types, obtaining a parent sample of 6875 galaxies. The final MFICA galaxy training sample, selected from the parent sample, consists of a random selection of 680 starburst galaxies, 1049 star-forming galaxies, 264 post-starburst galaxies, 554 green-valley galaxies, and 823 red galaxies. This gives us a training sample of 3370 galaxies.

2.3.2 MFICA quasar training sample

Quasar spectra were taken from the Schneider et al. (2010) SDSS DR7 Quasar Catalogue with redshifts calculated using the scheme of Hewett & Wild (2010). We selected quasars in the redshift range $0.25 < z < 0.5$, with mean per pixel spectral S/N > 8, and i-band absolute magnitudes $M_i < -23$, a magnitude brighter than the limit for inclusion in the catalogue. The M_i cut excludes most objects dominated by host-galaxy light, although we explicitly remove any remaining host-galaxy contamination in Section 2.4.2. Apart from a weak systematic change in emission-line equivalent width with luminosity (the Baldwin effect; Baldwin 1977), the spectral energy distributions (SEDs) of quasars do not depend significantly on bolometric luminosity or redshift (e.g. Temple et al. 2021). Thus, the SEDs of $M_i < -23$ quasars are representative of quasar SEDs in the full SDSS quasar catalogue. Dust-reddened quasars were removed using the ratio of the median fluxes between 4425–4775 and 3300–3600 Å in the quasar rest frame, requiring the ratio to be < 0.8. This eliminated the reddest 20 per cent of objects. Finally, we removed a small number of quasars with low-ionization BAL troughs. This gives us a final training sample of 2399 quasars.

2.3.3 Spectra acquisition and pre-processing

We use publicly available galaxy and quasar spectra from the SDSS seventh data release (DR7) (Abazajian et al. 2009), with OH molecule emission subtracted using the PCA prescription of Wild & Hewett (2005). We mask the 4279, 5578, 5894, 6300, and 6366 Å telluric emission lines and correct spectra for foreground Milky Way dust extinction using the O'Donnell (1994) dust extinction model and SDSS pipeline $E(B-V)$ values from Schlegel, Finkbeiner & Davis (1998). We shift all SDSS spectra to their rest frames using redshifts from Hewett & Wild (2010), and restrict the spectra to a rest frame wavelength range of 3300–5200 Å. We chose this wavelength range as it contains many stellar absorption features, including the 4000 Å and Balmer breaks, the H δ and higher order absorption lines, and important quasar features such as the blue continuum and broad H β emission. The 3300–5200 Å interval also includes the transition from AGN light domination at bluer wavelengths to galaxy light domination at redder wavelengths. To ensure that the full rest-frame wavelength interval is included in the spectra, we select objects in the redshift range $0.16 \leq z \leq 0.76$. Finally, we normalized the spectra using the mean of the fluxes in the 3300–5200 Å region.

2.4 Generating MFICA components

For the generation of positive components, we use an exponential prior as defined in Allen et al. (2013). When the positivity needs to be dropped, we use the Laplace prior defined in Allen et al. (2013).

2.4.1 Generating galaxy components

Using MFICA, we first derive galaxy components that encode the light emitted by stars and gas in a galaxy – a task shown by Kaban et al. (2005), Lu et al. (2006), and Allen et al. (2013) to be feasible with MFICA. In principle, a large number of components can be generated to allow reconstruction of galaxy spectra essentially perfectly. However, this may produce components that are degenerate with features in the quasar components and cause overfitting of the spectra. We thus limit the number of galaxy components we generate to mitigate this possibility.

Using components representing young, intermediate-age, and old stars, we expect to be able to reconstruct the stellar continuum and mean nebular emission of all galaxies to a sufficient extent for our purposes. Additional components are then derived to describe the variations in emission-line ratios. The PCA classifications of our training sample galaxies provide prior information on their stellar populations, which we can use to simplify the galaxy component creation.

We start by running MFICA on the sample of red galaxies. Red galaxies are composed of old stars with a low variance between spectra. We find that one positive component can reconstruct the spectra of most red galaxies to a level sufficient for the purposes of this paper. Given the similarity of the component to the mean of the red galaxy spectra in the training galaxy sample, we simply use the mean red galaxy spectrum as the first component. This component, shown in the top panel of Fig. 1, is labelled the K component as it resembles a K-giant star spectrum.

Next, we apply MFICA to the PSBs in the training sample. The spectra of PSBs are the next simplest, containing intermediate-age and old stars. We run MFICA looking for two positive components, including the previously determined K component. The MFICA analysis returns the component shown in the middle panel of Fig. 1, with signatures of A-type stars such as strong Balmer absorption lines. This component is labelled the AF component.

The remaining galaxy classes are composed of mixtures of young, intermediate-age, and old stars. We run MFICA on the entire training sample of 3370 galaxies, looking for six positive components, including the previously determined K and AF components. The first of the new components, shown in the bottom panel of Fig. 1, adds active star formation and O/B stars. However, it does not take the shape of an O/B type stellar spectrum due to the positivity constraint of our MFICA analysis, which forces the component to infill Balmer absorption and reduce spectral signatures of A and F stars.

The remaining three components, shown in Fig. 2, adjust emission-line ratios and are labelled GAL4-6. We generate an additional corrective component, GAL7, to improve emission-line reconstructions by altering velocity widths. This component was created by running MFICA on all the galaxy spectra with the contribution of the previous components subtracted, and the positivity constraint dropped.

2.4.2 Generating quasar components

We leverage the ability to pre-specify the form of components in the MFICA code to generate quasar components statistically independent of MFICA galaxy components. To account for contamination of the quasar spectra by light from the host galaxies, we first run MFICA on the quasar training sample with the K component fixed as the first component. We extract seven positive quasar components free of old stellar population signatures, such as the Ca II (H&K) and Mg I λ5175 absorption lines. The K component contributes an average of ~ 7 per cent by light to each quasar spectrum. We subtract the contributions

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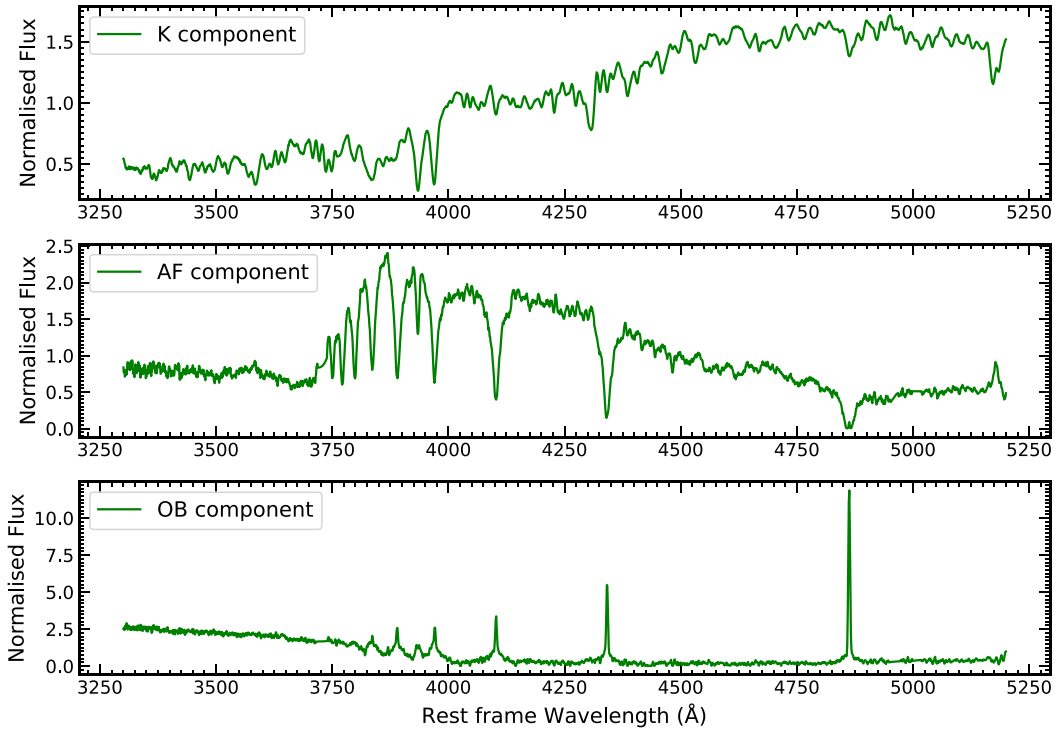


Figure 1. The MFICA galaxy components (f_λ). From top to bottom: the K component, AF component, and OB component.

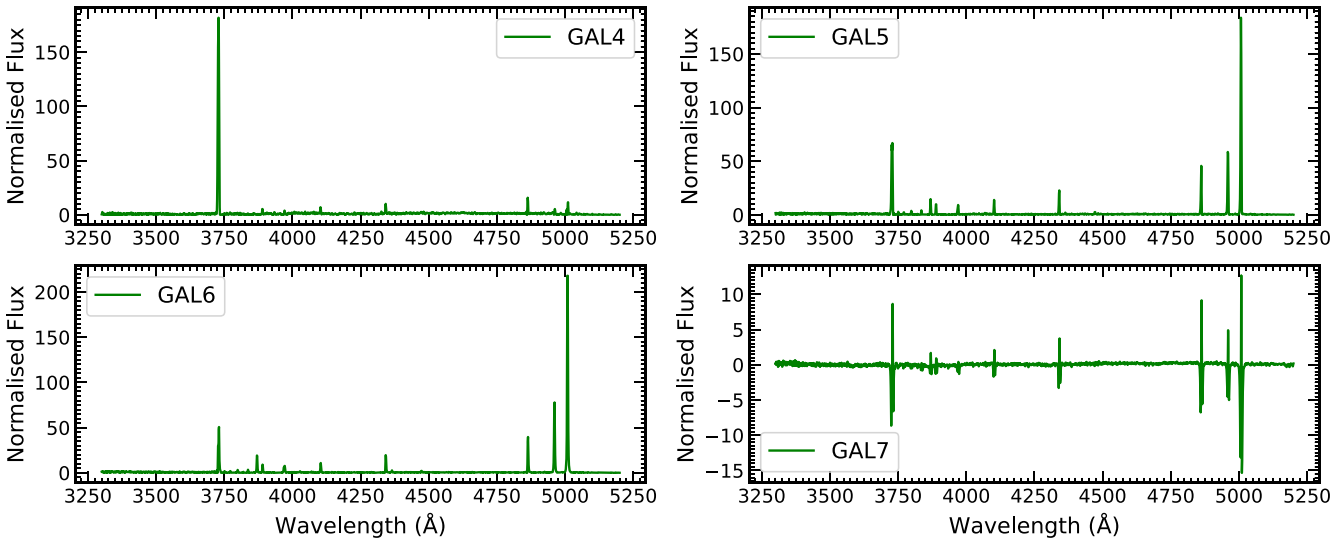


Figure 2. The MFICA emission-line components (GAL4-GAL6) and the corrective component (bottom right; GAL7).

of the K component from each spectrum, giving us training spectra free of old stellar population light. This step impacts the measured stellar mass of the host galaxies, and is slightly sensitive to the input quasar sample and the wavelength range of the components. We test our results with and without this step, and note where this introduces additional uncertainties to our results.

Next, we run MFICA on the K-subtracted quasar spectra with the AF component fixed instead. The average fractional contribution of the AF component to these spectra is ~ 2 per cent. Despite this small contribution, we subtract the AF contribution from the quasar spectra to minimize the presence of features such as strong Balmer absorption lines. We choose not to repeat the host galaxy subtraction with the OB component, owing to its similarity to an AGN continuum. Based on the average K and AF contributions however, the average OB contribution is expected to be ~ 1 per cent (see Fig. 5). This contribution is smaller than the error on the quasar components.

Finally, we need to ensure the QSO components are not contaminated by nebular emission from H II regions, as this could affect our ability to recover the galaxy stellar population, due to the correlation between H II region nebular emission and stellar continuum shape. To do this, we first remove the continuum of the combined quasar+galaxy spectra using a 61 pixel width median filter, which effectively isolates the narrow emission-lines. We then

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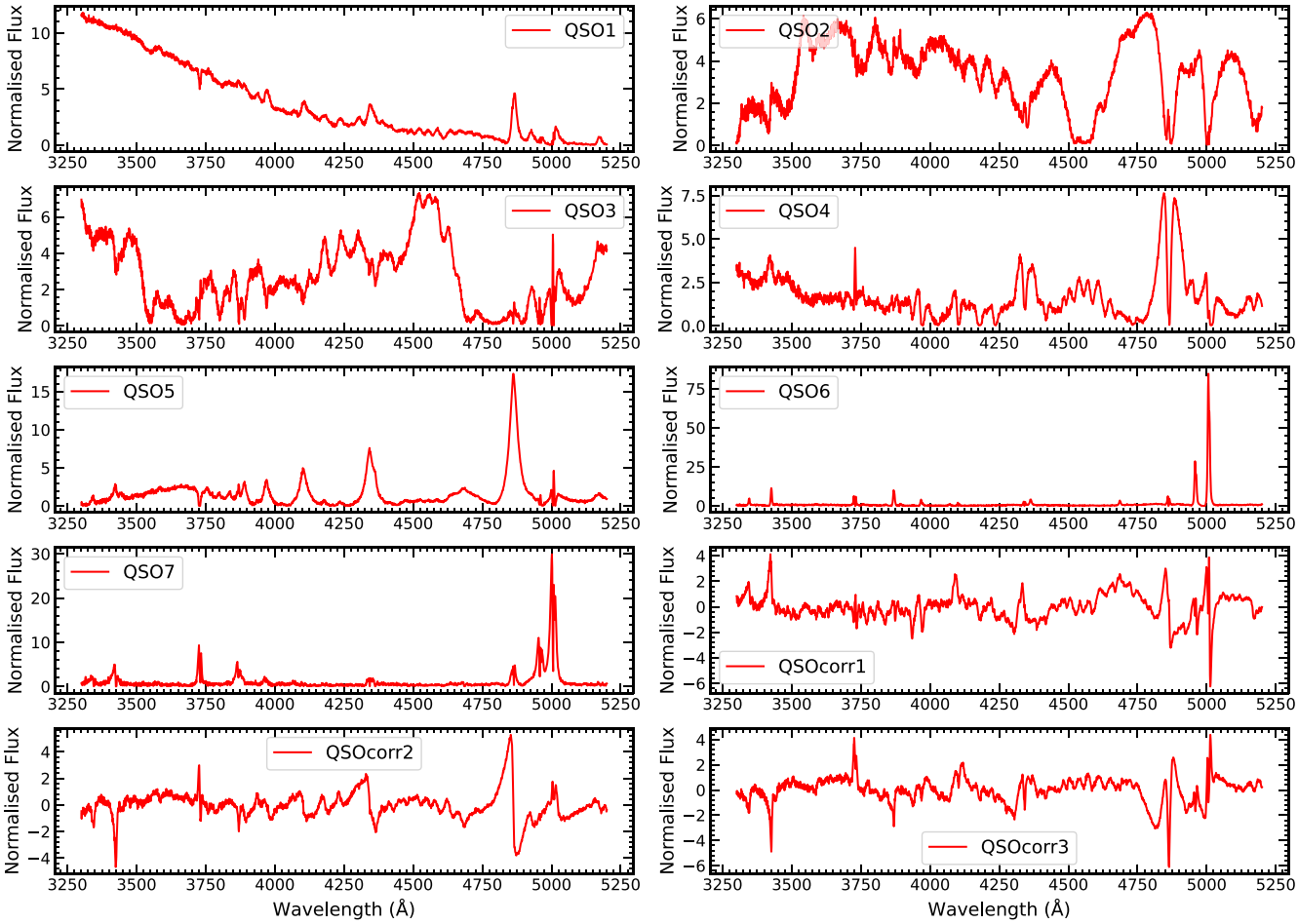


Figure 3. The MFICA quasar components (f_λ), after subtraction of host-galaxy contamination. The first seven components (QSO1–7) are the main positive components. The remaining three (QSOcorr1–3) are the corrective components generated with no positivity constraint.

run a new MFICA decomposition on the median-filtered spectra, fixing components GAL4, 5, and 6 and generating one further component (see e.g. Allen et al. 2013). This new component isolates the broader and often asymmetric AGN narrow-line-region lines. We then subtract the GAL4, 5, and 6 contributions from the K- and AF-subtracted quasar spectra and rerun the MFICA to generate galaxy-free QSO components.

Our emission line subtraction routine results in nearly all the [O II] λ3727, 3729 emission being assigned to the MFICA galaxy components, while GAL4, 5, and 6 contribute to the narrowest component of the [O III] λ4959, 5007 emission. The quasar components reproduce the higher ionization narrow emission (e.g. [Ne III] λ3870 and [Ne V] λ3427), the significantly broader velocity-width contributions to most of the lines and the blue-asymmetric outflow components (particularly for [O III] λ4959, 5007) that are common in quasar spectra (e.g. Mullaney et al. 2013). The average subtracted flux from GAL4, 5, and 6 is similar to the average flux of these components in starburst galaxies.

The seven MFICA quasar components obtained are shown in Fig. 3, labelled QSO1 to QSO7. These components reconstruct the galaxy-subtracted quasar spectra with an average root-mean-squared (rms) error of 7.4 per cent with respect to the quasar continuum. We generate three additional corrective components (with the positive prior dropped) from quasar-component-subtracted quasar spectra to improve the reconstruction of small residual features. The inclusion of these three corrective components reduces the average rms error to 7.3 per cent.

3 EXAMINATION OF GALAXY COMPONENTS

3.1 Validating the galaxy components

We test the effectiveness of the MFICA galaxy components by fitting the parent sample of galaxies with the linear combination of MFICA galaxy components. We obtain the best-fitting linear combination:

$$f_{\lambda,\,gal} = \sum_{i=0}^{7} w_{gal,i} \cdot e_{gal,i}, \tag{3}$$

where $e_{gal,i}$ are the galaxy components, and $w_{gal,i}$ the weights assigned to each component, by minimizing χ^2 using the error array provided by SDSS. The χ^2 (or weighted least-squares) fitting is handled by the Python package LMFIT (Newville et al. 2015), using the default Levenberg–Marquardt least-squares minimizer, which also provides the error on, and covariance between, the components.

To assess the performance of the MFICA components, we fit all the galaxy spectra with the MFICA galaxy components, and then stack (i.e. take the mean of) all galaxy spectra of a given type as well as their best-fitting reconstructions. In Fig. 4, we show the

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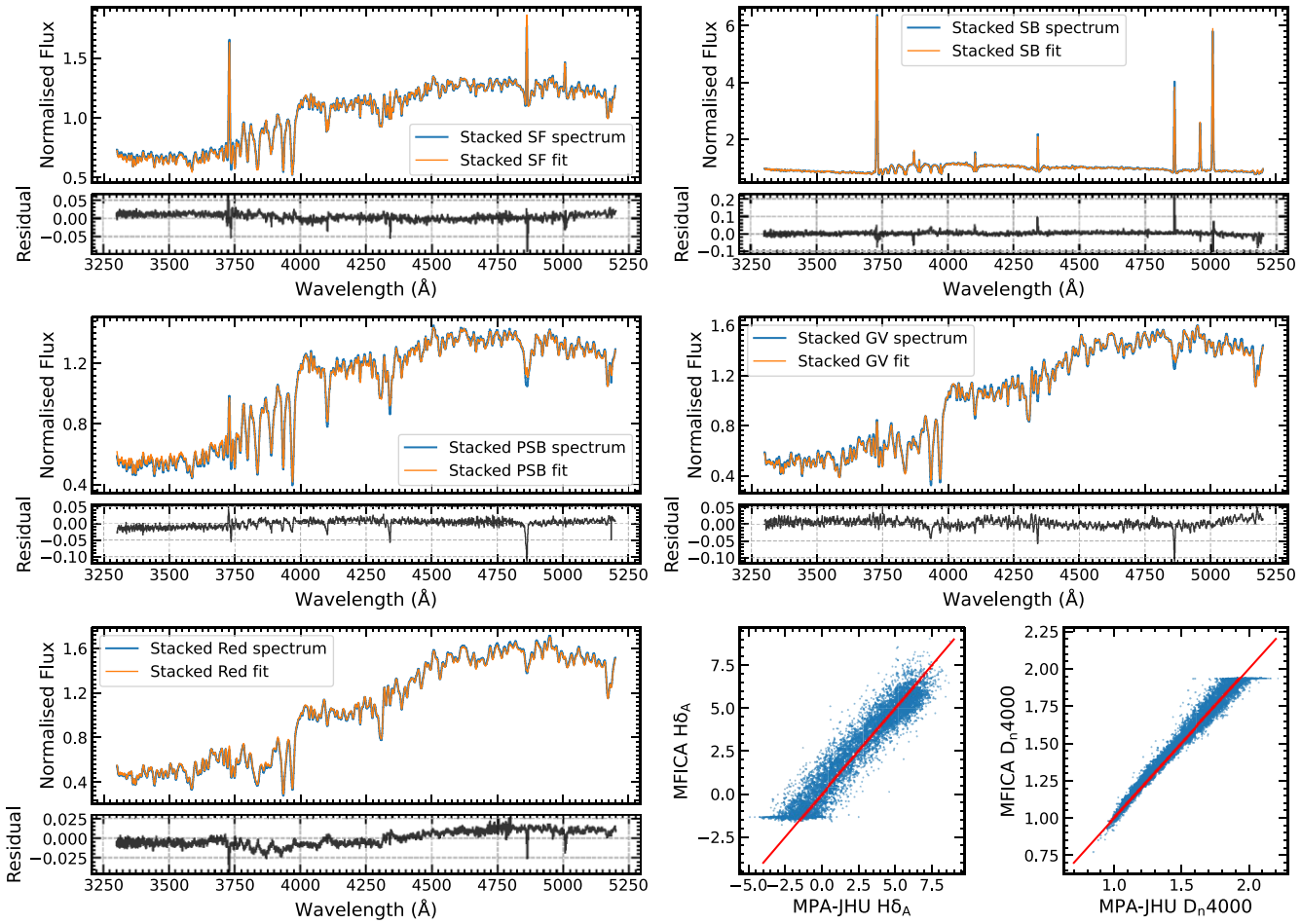


Figure 4. Stacked spectra and stacked reconstructions demonstrating the effectiveness of MFICA galaxy reconstructions. Upper panel of each subplot: the stacked (mean) spectra and stacked MFICA galaxy component fits of all spectra of a given galaxy spectral type are plotted in blue and orange respectively. Lower panel of each subplot: Difference (residual) between the galaxy spectra and fitted spectra stacks. The last two panels on the bottom right are comparisons of Hδ_A (left) and D$_n$4000 (right) measured from MFICA reconstructions on the *y*-axes, compared to their MPA-JHU catalogue values on the x-axes. The red line in these plots is the line of equality.

stacked galaxy spectra in orange and the stacked reconstructions in blue. Each panel shows the stacks for a given galaxy type and the residual difference between the stacked spectra and reconstructions. The MFICA galaxy components fit the galaxy spectra remarkably well given the small number of components, achieving a median reduced χ^2 of 1.02 and an average rms error of $\sim$ 1.2 per cent across all the stacks. Small residuals are present around Ca II (H&K), the Balmer emission lines, and the [O II] and [O III] doublets. While residuals could be further reduced by using more components, for this paper we deliberately chose to limit the number of components to avoid unwanted degeneracies or overfitting when fitting the host galaxies of quasars.

We also check if the MFICA can recover Hδ_A and D_n4000 measurements of the galaxy spectra. D_n4000 (as defined in Balogh et al. 1999) increases with the age of a galaxy's stellar populations (Kauffmann et al. 2003b). Hδ_A (as defined in Worthey & Ottaviani 1997) is largest for star-forming galaxies, or galaxies that have experienced a recent ($\leq$ 1–2 Gyr) burst in star formation (Kauffmann et al. 2003b). We use the emission-line free PCA components of Wild et al. (2007) to subtract emission lines from the MFICA reconstructions before calculating the Hδ_A and D_n4000 of the MFICA reconstructions. The last two panels in the bottom right of Fig. 4 show the Hδ_A and D_n4000 recovered by MFICA compared to those from the MPA-JHU catalogue.[1] The recovered values all lie close to the line of equality, proving that the MFICA galaxy components capture the properties of galaxies without bias.

3.2 Galaxy component fractions

The MFICA galaxy components resemble the spectra of stellar populations of different ages. The weight assigned to these components thus encodes information on the abundance of stellar populations in galaxies. To explore this, we define three quantities – f_K, f_{AF}, and f_{OB} – that represent the fractional contribution of each component to the total flux from these three components integrated across the 3300–5200 Å interval. That is, the fraction of the *i*th galaxy component is defined as

$$f_{gal,i} = \frac{\langle w_{gal,i} \cdot e_{gal,i} \rangle}{\left\langle \sum_{i=1}^{3} w_{gal,i} \cdot e_{gal,i} \right\rangle}, \quad (4)$$

[1]The hard cuts on D_n4000 and Hδ_A are caused by the use of a single red galaxy template, as opposed to an inherent limitation of MFICA.

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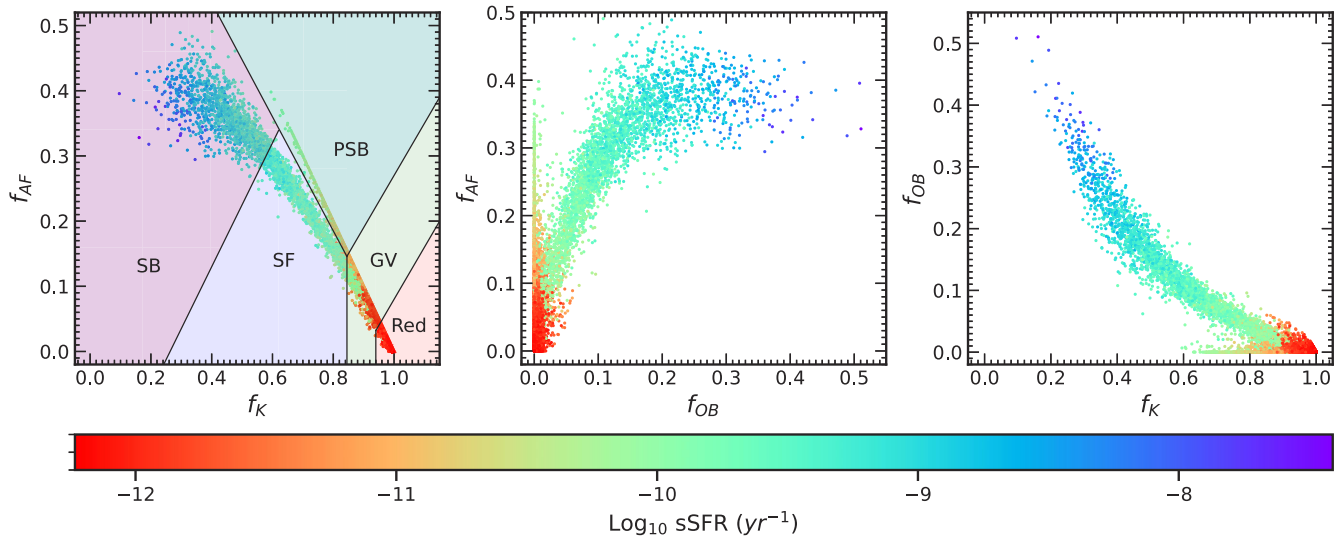


Figure 5. Scatter plots of galaxy component fractions, derived from fitting the parent galaxy sample with the MFICA galaxy components. Galaxy component fractions (dots) are colour coded by sSFR from Brinchmann et al. (2004). Left: K-component versus AF-component fraction. The boundaries in this panel demarcate the approximate regions for different galaxy types. These boundaries are used in quasar host galaxy classifications. Middle: OB-component versus AF-component fraction. Right: K-component versus OB-component fraction. In all three plots, the primary axis of variation is the sSFR sequence of normal galaxies, while the post-starburst galaxies form a distinctly separate population in MFICA space.

where $e_{gal,i}$ is the flux of the $i^{\rm th}$ galaxy component, $w_{gal,i}$ are the weights defined in equation (3), and $\langle\rangle$ represents the mean over the full 3300–5200 Å wavelength range. The sum in the denominator represents the sum of the fluxes assigned to the K, AF, and OB components only. For example, an f_K of 0.8 implies that 80 per cent of the reconstructed galaxy's continuum light comes from the K component.

The component fractions of the parent galaxy sample are plotted in pairs in Fig. 5, with objects colour-coded by their specific star formation rates (sSFRs) where the sSFR was measured using Hα luminosity, corrected for dust attenuation using the Balmer decrement (i.e. $i_{class} = 1$ or 2 in Brinchmann et al. 2004).[2] In Fig. 6, we focus on the middle panel of Fig. 5 and show how the shape of a galaxy spectrum changes as the component fractions change. These plots show that the components do encode the abundance of stellar populations. Objects with low f_{AF} and f_{OB}, but f_K close to unity, are red/quiescent galaxies with low sSFRs, dominated by old stellar populations. On the other hand, f_{AF} and f_{OB} are the highest (and f_K close to zero) for star-forming and starburst galaxies with abundant young stellar populations and high sSFRs. As f_{AF} and f_{OB} increase, and f_K decreases, the stellar populations become younger. As shown in the top right panel of Fig. 6, moving from low f_{AF} and f_{OB} weight (pink) to a high f_{AF} and high f_{OB} weight (cyan) results in the flattening of the galaxy spectra as their 4000 Å break strengths decrease and emission-line strengths increase.

Additionally, for the same f_K, galaxies that have experienced a recent burst of star formation have a higher f_{AF} and lower f_{OB}. This is expected for post-starburst galaxies with a dearth of short-lived ($\sim$ 100 Myr lifetime) O/B stars after a rapid burst and subsequent cessation in star formation, but abundant A/F type stars. Moving along this branch of post-starburst galaxies (bottom right panel of Fig. 6, from orange to black), the flattening of the spectral slope is accompanied by an increase in Balmer absorption line strength and swap from a 4000 Å to a Balmer break.

It is thus clear that galaxies with distinct SFHs occupy different regions in MFICA component fraction space, allowing MFICA to be used to select/classify galaxies with distinct SFHs. We define boundaries (shown in the left panel of Fig. 5) visually, noting that the location of these boundaries is somewhat arbitrary, apart from the positioning of the post-starburst galaxies where a strong bimodality is seen. The membership criteria based on these boundaries is as follows:

(i) starburst: ($f_{AF} < -0.875\,f_K + 0.885$) and ($f_{AF} > 0.95\,f_K - 0.25$);
(ii) star-forming: ($f_{AF} < 0.95\,f_K - 0.25$) and ($f_{AF} < -0.875\,f_K + 0.885$) and ($f_K < 0.845$);
(iii) post-starburst: ($f_{AF} > -0.875\,f_K + 0.885$) and ($f_{AF} > 0.8\,f_K - 0.53$);
(iv) green-valley: ($f_K > 0.845$) and ($f_{AF} < 0.8\,f_K - 0.53$) and (($f_K < 0.94$) or ($f_{AF} > 0.8\,f_K - 0.72$));
(v) quiescent: ($f_K > 0.94$) and ($f_{AF} < 0.8\,f_K - 0.72$).

We will use these boundaries and membership criteria to classify inactive galaxies and quasar host galaxies, based on their SFHs.

The distribution of the component fractions closely resembles the Hδ_A–D_n4000 diagram (Kauffmann et al. 2003a) and the PC1–PC2 diagram (Wild et al. 2007). All three diagrams encode information on the SFHs of galaxies, although the MFICA component weights are more directly related to the stellar populations of galaxies.

4 VALIDATING QUASAR SPECTRAL DECOMPOSITION WITH MOCK SPECTRA

In this section, we describe how we decompose spectra using the MFICA quasar and galaxy components and validate our approach by constructing and decomposing mock spectra. In Section 4.1, we describe the decomposition procedure. In Section 4.2, we describe the construction of mock spectra, and in Section 4.3, we assess the quality of the mock decomposition. Finally, in Section 4.4, we

[2]For the purposes of this plot only, we excluded galaxies with Hα equivalent width < 3 Å, due to contamination of this measurement for 'retired' galaxies (Cid Fernandes et al. 2010).

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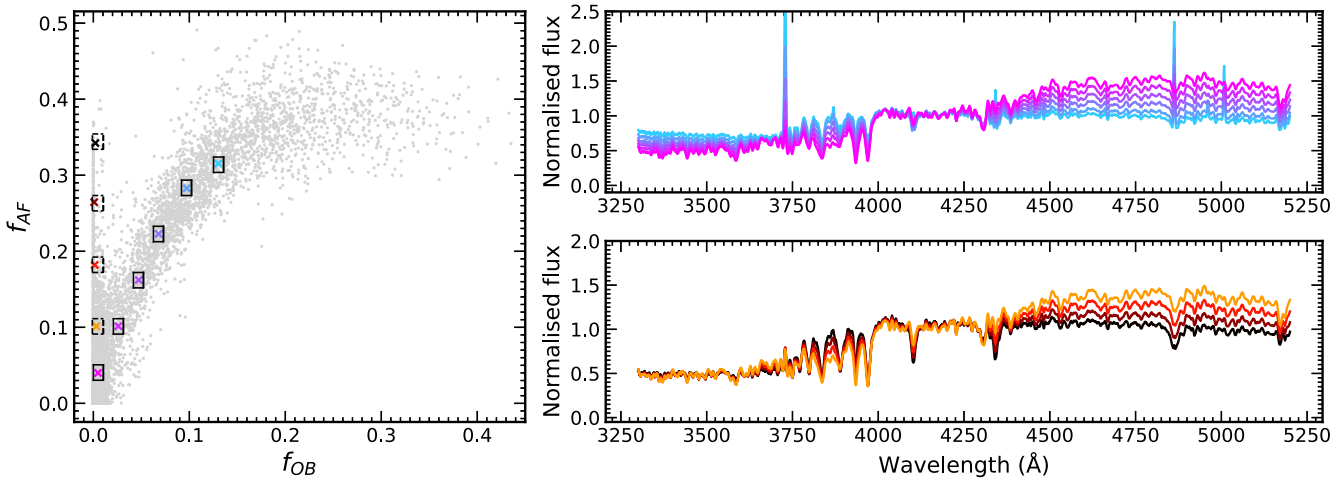


Figure 6. Demonstration of how a change in AF or OB component fraction corresponds to a change in shape of a galaxy spectrum. Left: AF-component versus OB-component fraction of the parent galaxy sample. The black boxes demarcate regions from where galaxy spectra were stacked. The solid (dashed) boxes correspond to the stacked spectra in the top (bottom) right panel. Top right: stacked spectra of different regions in the 'main-sequence' of galaxies in AF and OB component fractions. For each stack, the corresponding AF and OB component fractions are marked by the cross of the same colour. Bottom right: stacked spectra of different regions within the post-starburst branch in f_{AF} and f_{OB} space. All the spectra have been normalized to their mean value between 4100 and 4200 Å.

describe our technique to measure quasar host galaxy stellar masses. To determine the efficacy of the decomposition for SDSS spectra, we ask the following questions:

(i) How well can MFICA decomposition recover the fraction of the total light contributed by the host galaxy (f_{gal})?
(ii) How accurately can we recover the galaxy component fractions? In what limits can we accurately classify the host galaxies of quasars?
(iii) How does the recovery of component fractions vary as a function of (a) S/N of the mock spectra; (b) f_{gal} of the mock spectra?

4.1 Decomposing quasar spectra with MFICA

In order to decompose a quasar spectrum into its quasar-only and host galaxy parts, we fit it with a linear combination of the MFICA galaxy and quasar components simultaneously:

$$f_{\lambda,total} = \left(\sum_{i=0}^{7} w_{gal,i} \cdot e_{gal,i}\right) + \left(\sum_{j=0}^{10} w_{qso,j} \cdot e_{qso,j}\right), \quad (5)$$

where $e_{gal,i}$ is one of the seven galaxy components, $e_{qso,j}$ is one of the 10 quasar components, and $w_{gal,i}$ and $w_{qso,j}$ are the weights assigned to the galaxy and QSO components.

We fit the model to spectra using LMFIT, following a three-step procedure:

(i) First, all galaxy and quasar components, barring the corrective components, are fit together. Each component is assigned an initial weight of 0.1, with a minimum of 0 as the components are positive.
(ii) Next, we repeat the full fit with the inclusion of galaxy and quasar corrective components. We assign an initial value of zero to the corrective components and allow the quasar corrective weights to vary within $+/-3\sigma$ of their known distributions, as determined from the quasar training sample. All other components are fixed at their previously obtained values.
(iii) In the final step, we drop any components whose weights are less than 0.0001, simplifying the model fit, which in turn allows LMFIT to estimate errors for all component weights (Newville et al. 2015).

LMFIT returns errors on the component weights ($w_{gal,i}$ and $w_{qso,j}$) along with a covariance matrix (Newville et al. 2015). We use error propagation to obtain errors on the galaxy component fractions (f_K and f_{AF}) from the errors on the galaxy component weights ($w_{gal,i}$), following equation (4).

In Section 5.2, we calculate the percentage of quasar host galaxies in each of the five SFH categories shown in Fig. 5. To incorporate uncertainties into this calculation, fully accounting for correlations between the components, we resample the component fractions and re-calculate the fraction of quasar host galaxies in each category 10 000 times using the covariance matrices. The mean and standard deviation of the distributions are reported as the final percentages and errors for each SFH category.

4.2 Creating mock spectra

While we assume that a quasar spectrum can be decomposed by fitting it with linear combinations of MFICA components, the success of such an approach is not guaranteed. We cannot, however, test our assumption using real quasar spectra, as properties of their host galaxies are not known a priori. We thus create mock spectra, where input parameters are known, to test the MFICA decomposition.

We adopt the following recipe to create mock spectra that resemble real SDSS DR7 spectra:

(i) For the host galaxy, we construct high S/N composite spectra by taking the mean of all galaxy spectra in each spectral class in our parent sample: star-forming (SF), starburst (SB), post-starburst (PSB), green-valley (GV), and quiescent (red). These composites are not perfectly reconstructed by the MFICA components (see Fig. 4), and are thus out of MFICA parameter space, ensuring that decomposition of mocks is non-trivial.
(ii) For the quasar, we use MFICA reconstructions of the quasar training sample. This ensures the mocks represent the full diversity of spectra seen across the quasar population, and also that the mock quasar spectra are free from host galaxy contamination.

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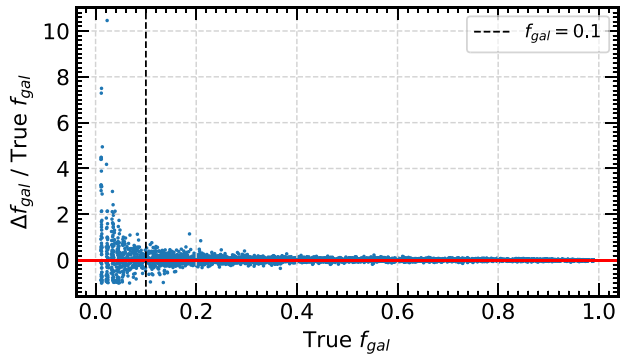


Figure 7. Results from mock spectra to demonstrate the fractional accuracy of host galaxy fraction (f_{gal}) recovered from the MFICA quasar spectral decomposition, as a function of the true f_{gal}. The dashed line represents the lower limit of $f_{gal} = 0.1$ below, which the recovered f_{gal} is no longer accurate.

(iii) We combine random pairs of quasar reconstructions and galaxy composite spectra using a random value (between 0 and 1) for the fraction of light coming from the host galaxy (f_{gal}).

(iv) We add noise to our mock spectra using an error array from a random SDSS quasar spectrum (MJD = 51817, plate = 418, fibre = 566), scaled to achieve a specific median S/N between 3300–5200 Å. We resample the fluxes of the mock spectra from a normal distribution with σ equal to the scaled error array.

4.3 Results from decomposing mock spectra

First, we test the recovery of the mean f_{gal}, measured over the full $3300 - 5200$ Å interval. For each host galaxy composite, we create 1000 mock spectra, assuming a median S/N ratio of 10 (close to the median of the SDSS DR7 quasar catalogue ~ 10.8), with f_{gal} ranging from 0 to 1, resulting in a total of 5000 mock spectra. We decompose each mock using the procedure outlined in Section 4.1. In Fig. 7, we plot the fractional difference in the recovered f_{gal} compared to the true f_{gal}, as a function of the true f_{gal}. We find that MFICA can accurately (rms scatter in $\Delta f_{gal}/f_{gal}$ of 0.09) recover f_{gal} for $f_{gal} \geq 0.1$. Below this limit, the recovery of f_{gal} worsens significantly.

Next, we quantify the S/N ratio and f_{gal} limits above which galaxy component recovery is accurate. The dependence of component fractions on f_{gal} and S/N ratio are shown in Appendix A. We create 2000 mock spectra for each host galaxy composite for this task, adopting a random f_{gal} between 0.1 and 0.9. In Fig. 8, we plot the component fractions recovered after decomposition as coloured dots, with contours representing the 68 and 95 per cent confidence intervals for S/N= 10 mock spectra. We find that at a median S/N ratio ≥ 10 and an $f_{gal} \geq 0.2$, the 95 per cent confidence contours for the decomposed galaxy component fractions do not overlap, and the frequency of outliers is minimized. In real data, these outliers are characterized by the similarity between the rising continuum fluxes of the OB component and quasars (e.g. Bessiere et al. 2017), a known challenge for previous decomposition techniques (Matsuoka et al. 2015; Wei et al. 2018). Above the S/N ≥ 10 and $f_{gal} \geq 0.2$ limits, f_{AF} and f_K show low mean systematic offsets of 0.36 and 0.42 per cent and are well constrained to within an rms spread of 3–4 per cent with respect to their dynamic ranges. We thus consider MFICA decomposition to be accurate above a S/N ratio of 10 and $f_{gal} \geq 0.2$.

We also tested the decomposition using real galaxy spectra rather than high S/N composites. We find that we are able to recover the demographics of the input mock galaxies accurately.

4.4 Stellar mass for quasar host galaxies

In what follows, we compare our results to a sample of inactive galaxies matched in galaxy stellar mass, which requires an estimate of the stellar mass of the quasar host galaxies. The standard method to estimate the stellar mass of a galaxy is to fit template stellar population models to observed spectra or photometry, to obtain a mass-to-light ratio which is then scaled to the galaxy's luminosity. We tried this standard approach of least squares fitting the reconstructed host galaxy continuum spectra with the 'stochastic burst' library of Bruzual & Charlot (2003) stellar population models, masking the primary nebular emission lines, following the philosophy of Kauffmann et al. (2003a) and Salim et al. (2005). We tested this method's efficacy for our work by fitting spectra of MPA-JHU galaxies and comparing the resultant stellar masses to the fibre stellar masses in the MPA-JHU catalogue. The MPA-JHU stellar masses were estimated from SDSS fibre aperture photometry using the same standard model fitting approach.[3] We find that the 5200 Å red wavelength limit used for our components is insufficient to fully constrain the galaxy dust attenuation. This leads to a small (~ 0.1dex) overestimate in the stellar masses compared to the MPA-JHU fibre stellar masses, as the model fits tended towards the median of the assumed dust attenuation prior.

Therefore, to better use natural prior correlations between dust attenuation, stellar populations and stellar mass in galaxies, we instead trained a random forest regressor (from SCIKIT-LEARN, with 100 estimators) on fibre stellar masses of SDSS DR7 spectroscopically classified galaxies in the MPA-JHU catalogues. We normalized these photometric stellar masses by the luminosity of the galaxy spectrum measured over the $3300 - 5200$ Å wavelength range to get a mass-to-light (M/L) ratio. The random forest regressor was trained on f_{AF} and f_K (obtained from reconstructing galaxy spectra with MFICA) as inputs, and asked to predict M/L. Re-normalizing the resulting M/L by the measured galaxy spectrum luminosity gives the final galaxy stellar mass, normalized to the fibre aperture photometry.[4] Random forests are extremely capable at learning smooth non-linear relationships between variables, as is the case between f_{AF}-f_K and M/L.

We selected the training and test galaxies for the random forest to have $0.16 \leq z \leq 0.76$, median S/N> 8, $i_{class} \leq 2$ (Brinchmann et al. 2004) and secure redshifts ($z_{warn} = 0$). A small number of objects were removed with measured $D_n(4000) < 0$. This results in 109 180 galaxies, which were randomly split into 70 per cent training and 30 per cent test samples. The stellar masses predicted for the test galaxies have negligible systematic offset compared to the MPA-JHU fibre stellar masses, and an rms scatter of 0.084 dex, with no systematic deviations for different galaxy spectral types (as defined in Section 3.2. We also verified that different galaxy training sets did not result in different stellar masses for the quasar hosts.

At the redshifts applicable to this work ($z > 0.16$), the difference between galaxy stellar masses measured from SDSS fibre and model

[3]Available from https://wwwmpa.mpa-garching.mpg.de/SDSS/DR7/Data/stellarmass.html

[4]We note that the SDSS 'fibre aperture' photometry does not have the same normalization as the SDSS spectra for galaxies, and our method accounts for this difference.

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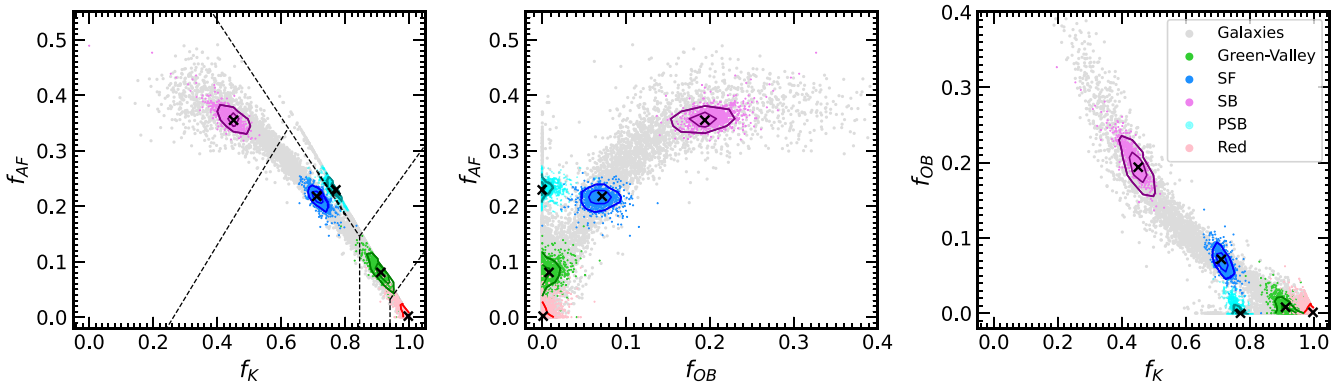


Figure 8. Scatter plots of the galaxy component fractions from decomposing mock quasar+galaxy spectra with a median S/N of 10 and $0.2 \leq f_{gal} \leq 0.9$, i.e. in the limits where the decomposition is accurate. The crosses represent the true component fractions and contours represent the component fractions recovered from decomposing the mock quasar+galaxy spectra. The contours represent 68 and 95 per cent confidence intervals. Left: K-component versus AF-component fraction. Middle: OB-component versus AF-component fraction. Right: K-component versus OB-component fraction.

magnitudes in the MPA-JHU catalogue is 0.5 dex, independent of redshift. This is because at these redshifts the loss of light is dominated by atmospheric seeing, rather than the fraction of galaxy covered by the physical extent of the fibre. When comparing to other work, we add a constant 0.5 dex aperture correction to our stellar masses to estimate the total host galaxy mass. Aperture corrections depend on the size and structure of the galaxy, and by using the constant 0.5 dex correction, we are implicitly assuming the inactive control sample has a similar galactic size distribution to the quasar host galaxies, which may not be entirely correct (Li et al. 2021). However, this aperture correction is not required for our analysis or comparison with the inactive control sample.

5 RESULTS

In this section, we apply our MFICA decomposition code to study the stellar populations of quasar host galaxies. We select quasars from the SDSS DR7 quasar catalogue (Schneider et al. 2010) in the redshift interval $0.16 \leq z \leq 0.76$. For determining the stellar populations of our quasar hosts, we limit this initial sample of 19 358 quasars to those with an $f_{gal} \geq 0.2$, a median S/N ratio ≥ 10 (see Section 4.3), and $\chi^2_\nu \geq 2.0$ which results in 3376 quasars. This sample of reliably decomposed quasars has a bolometric luminosity range of $44.7 \leq \log_{10}(L_{bol}/\mathrm{ergs}^{-1}) \leq 47.0$.

The left hand panel of Fig. 9 shows an example of a decomposed quasar spectrum with the reconstructed host galaxy and AGN spectra. There are no significant features in the residuals, and this is typical of the full sample. The majority of quasar spectra are well fit with a median reduced χ^2 (χ^2_ν) of 1.07. The right hand panel of Fig. 9 shows the recovered component fractions of the quasar host galaxy, with error bars calculated as described in Section 4.1.

Fig. 10 shows the distribution of host galaxy fractions (f_{gal}) of our sample of quasars, where f_{gal} is measured over the full $3300 - 5200$ Å interval. Quasars are colour-coded by their bolometric luminosities (L_{bol}) as measured by Shen et al. (2011). The typical f_{gal} decreases with increasing redshift and L_{bol}. This trend may be explained by invoking equation (7) of Temple et al. (2021), which relates the galaxy and quasar luminosities to host galaxy fractions. L_{bol} increases with increasing redshift as the SDSS quasar catalogue is flux-limited, including only those objects with $m_i < 19.1$ (Schneider et al. 2010), which at the highest redshifts correspond to the highest luminosities. However, per the SDSS-calibrated model of Temple et al. (2021) the luminosity of galaxies rises slower than the rise in quasar luminosity, resulting in a decrease in f_{gal} with redshift.

5.1 Quasar host stellar masses and control sample selection

The left panel of Fig. 11 shows the distribution of estimated fibre stellar masses for the quasar host galaxies. Quasars (with $f_{gal} \geq 0.2$) are preferentially hosted by massive galaxies with a median fibre stellar mass (i.e. based on light collected in the SDSS 3 arcsec fibre) of $10^{10.8}$ $M_\odot$ and an rms deviation from median of 0.3 dex. This increases to a total stellar mass of $\sim 10^{11.3}$ $M_\odot$ after accounting for a fixed 0.5 dex aperture correction (Section 4.4).

The stellar populations of galaxies depend on stellar mass: galaxies with higher stellar masses have older stellar populations on average. Previously derived differences between the stellar populations of quasar hosts and inactive galaxies may be due to differences in their stellar masses (Trump et al. 2013). We control for the effect of stellar mass when comparing quasar host and inactive galaxies by constructing a stellar-mass matched control sample of galaxies. For each quasar host, we chose two galaxies from the MPA-JHU catalogue with the closest fibre stellar mass. The distribution of fibre stellar masses and redshifts for the 3376 quasars and 6752 control galaxies is shown in Fig. 11. At the high mass end ($\sim 10^{11.5} M_\odot$), the fibre stellar mass distributions for the quasar hosts and the control sample do not match exactly. This mismatch is attributable to the larger uncertainties on the stellar masses of the quasar host galaxies which, combined with the steep fall-off in the high-mass end of the galaxy stellar mass function, leads to scatter beyond the high mass limit of the control galaxy sample. The number of objects without exact matches is minor, and this effect will not significantly impact the results.

5.2 Quasar host-galaxy properties

In Fig. 12, we compare the AF and K-component fractions (f_{AF} and f_K respectively) for the quasar hosts and the control sample. Our sample of quasar host galaxies have a wide range of f_{AF} and f_K. The quasar hosts, however, have generally higher f_{AF} and lower f_K than the control sample, implying that quasar host galaxies have younger stellar populations than stellar-mass-matched control galaxies. The younger stellar populations of quasar hosts can also be inferred from the median f_K and f_{AF}. The median quasar host is a star-forming galaxy, and the median control galaxy is quiescent. The younger

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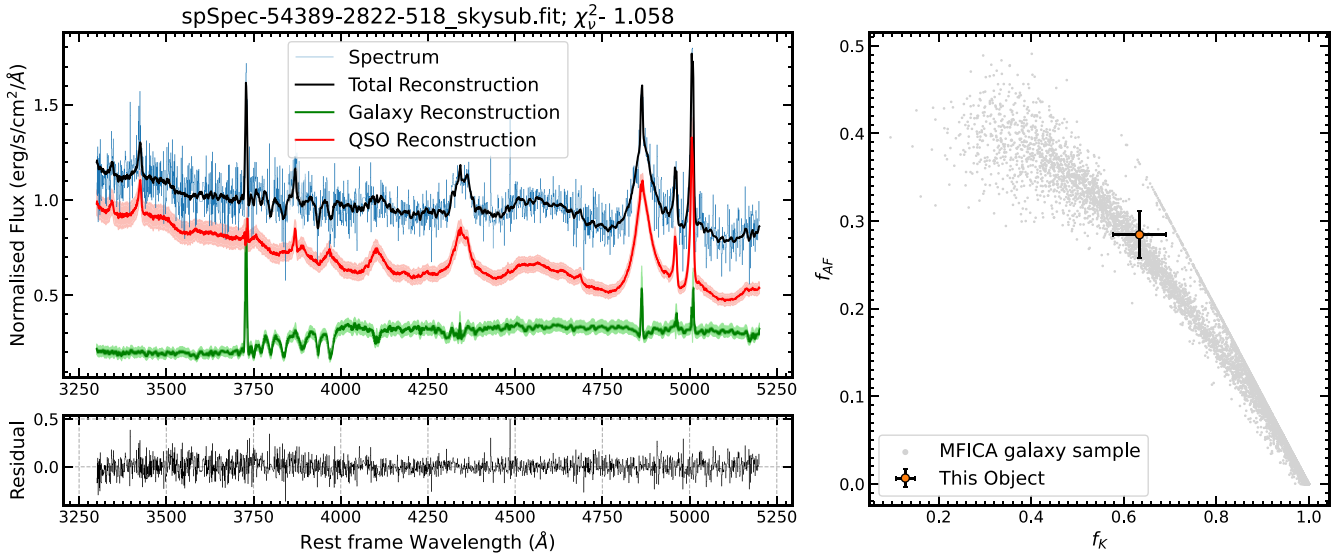


Figure 9. The left panel shows an example of the decomposition of a quasar spectrum. Top: the quasar spectrum (blue), best-fitting MFICA model (black), reconstructed host galaxy (green), and reconstructed quasar (red). Bottom: the difference between the quasar spectrum and the best-fitting model. The right panel shows the location of this quasar host galaxy (orange dot) in the K-component versus AF-component fraction space. The error bars are determined from the errors on the weights from LMFIT (see Section 4.1). The grey dots represent the distribution of the parent galaxy sample.

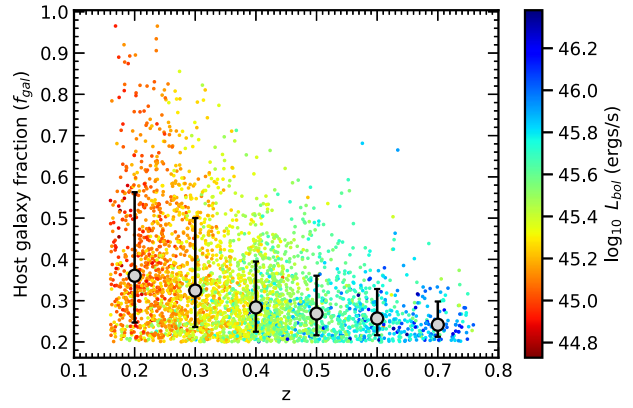


Figure 10. Variation in host galaxy fraction as a function of redshift for quasars with $f_{gal} \geq 0.2$, median $S/N \geq 10$, and $\chi^2_\nu < 2.0$. The colours correspond to the bolometric luminosity of the quasars (Shen et al. 2011). The grey filled circles represent the median f_{gal} in redshift bins of width 0.1, and the bars represent the 16th and 84th percentiles.

stellar populations of the quasar host galaxies can also be seen in Fig. 13, which compares the stack (mean) of all the reconstructed quasar host galaxy spectra, and the stack of all the reconstructed control galaxies. The stacked quasar host spectrum has a smaller D_n4000, indicative of a young stellar population and active star formation.

Fig. 14 shows the percentage of different galaxy types in the quasar host and control samples. The height of each bar corresponds to the percentage of either the ($f_{gal} \geq 0.2$) quasar sample or the control sample classified as a particular galaxy type (see the left panel of Fig. 5). The dot-filled bars represent the percentages for the quasar sample, while the line-filled bars represent the percentages for the control sample. The information in this plot is also summarized in Table 1. The error bars in Fig. 14 represent the maximum and minimum percentage for each host galaxy type, while Table 1 shows the true errors in the percentages.

We find that the biggest class of quasar host galaxies are star-forming galaxies (∼47 per cent; ∼53 per cent when including starburst galaxies), with a significant number of post-starburst (∼25 per cent) and green-valley hosts (∼18 per cent). On the other hand, the vast majority (∼73 per cent) of our stellar mass matched control galaxies are red/quiescent, ∼12 per cent are star-forming galaxies, ∼2 per cent are starburst, ∼11 per cent are green-valley galaxies and <1 per cent are post-starburst galaxies. These results imply that the stellar populations and recent SFHs of quasar host galaxies and control galaxies are distinct.

Apart from a higher fraction of star-forming quasar hosts, we also see that quasar hosts include a larger fraction of post-starburst galaxies than the control sample. While ∼25 per cent of all quasar hosts are post-starburst galaxies, <1 per cent of our control sample are post-starburst galaxies, which is consistent with previous results (Wild et al. 2016; Rowlands et al. 2018). Our analysis shows that post-starburst galaxies are 28 ± 1 times more common amongst quasars than in the control galaxies of the same stellar mass.

The excess of post-starburst galaxies in quasar hosts is substantially driven by the large number of quiescent control galaxies. Therefore, we can alternatively represent the excess of post-starbursts by normalizing by the total number of star-forming galaxies. For the same number of star-forming galaxies, it is 7.3 ± 0.4 times more likely for a quasar to have a post-starburst host than it is for an inactive galaxy to be a post-starburst galaxy.

5.3 Redshift evolution and stellar mass uncertainties

The comparatively larger fraction of quiescent galaxies in the control sample could be dictated by their lower redshifts, as the fraction of quiescent galaxies increases over time (Noeske et al. 2007; Ilbert et al. 2015). On the other hand, due to cosmic down-sizing, the fraction of massive quiescent galaxies does not change substantially over the redshift ranges probed here (Ilbert et al. 2013; Muzzin et al. 2013; Rowlands et al. 2018; McLeod et al. 2021; Weaver et al. 2023). We are unable to control for redshift, as the redshift distributions of SDSS quasars and galaxies are entirely different

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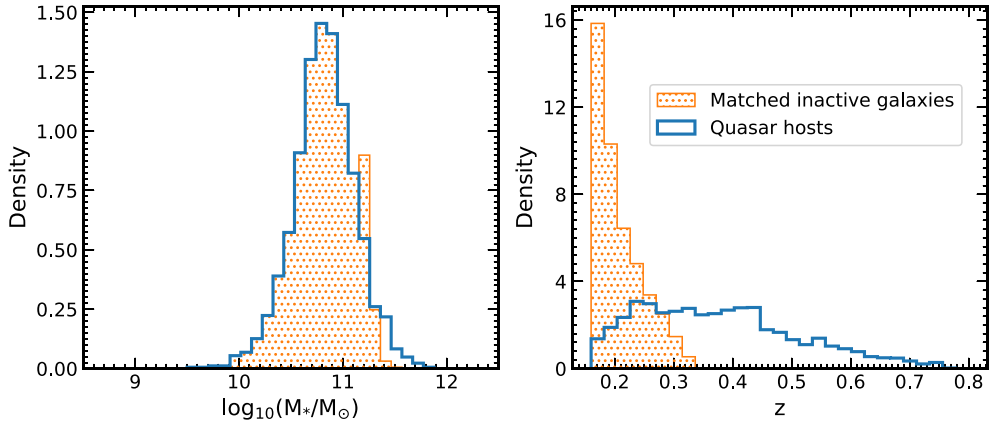


Figure 11. Left: the normalized histograms of the stellar masses for the quasar host galaxies (solid blue) and the control sample (dotted orange), measured in the 3 arcsec SDSS fibre. Right: the redshift distribution of the quasar host galaxies (solid blue) and the control sample (dotted orange).

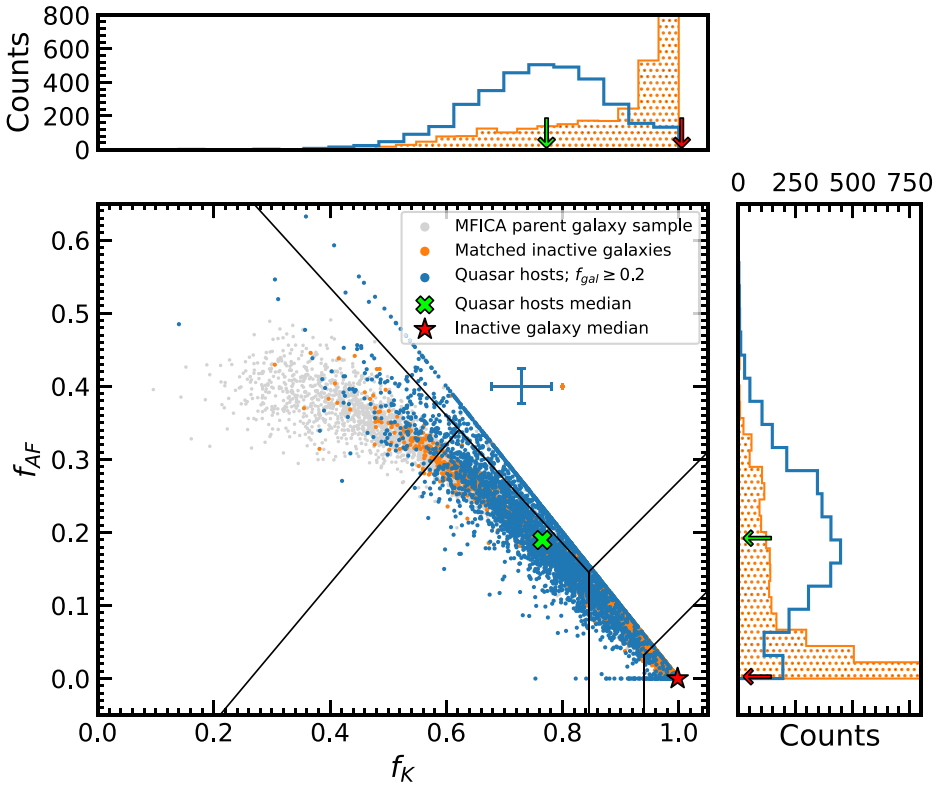


Figure 12. K-component fraction versus AF-component fraction for quasar host galaxies with $f_{gal} \geq 0.2$ (blue dots), and stellar-mass matched control galaxies (orange dots). The light-grey dots represent the distribution of the MFICA parent galaxy sample (for visual-aid only). The green-cross is the median of the weights of all quasar host galaxies, and the red star is the median of the weights of all control galaxies. The error bars represent the mean of the component fraction errors for the decomposed quasar host galaxies (blue and larger) and the control sample (orange and smaller). Right and top: projected histograms of AF and K-component fractions for quasar host galaxies (blue solid line) and the control sample (orange dotted histogram). The control sample bins with highest K-component fraction and lowest AF-component fraction have been clipped at 800 counts – there are an additional 3740 galaxies in the final AF-component fraction bin, and 3929 in the final K-component fraction bin.

(Richards et al. 2002; Strauss et al. 2002). Therefore, we checked if redshift differences drive our results by restricting the quasar sample to $z < 0.3$, close to the maximum redshift of the control sample. The fraction of star-forming quasar host galaxies (47.9 ± 1.1 per cent at $z < 0.3$) does not change within errors, whereas the fraction of quiescent host galaxies rises slightly from 4.6 ± 0.2 to 6.2 ± 0.4 per cent. Quiescent galaxies, however, still dominate the control galaxy sample even as their fraction drops from 73.92 ± 0.09 to 64.9 ± 0.2 per cent. This drop occurs due to the lower stellar masses of the quasar hosts (and thus control galaxies) at $z < 0.3$, owing to lower luminosities and consequently (at fixed Eddington ratio) lower black hole masses. These results show that the most important variable is stellar mass, and our conclusion that quasar host galaxies have younger stellar populations than the control galaxies holds firm.

Similarly, the post-starburst fraction in ordinary galaxies increases with redshift, although not particularly strongly for the highest mass

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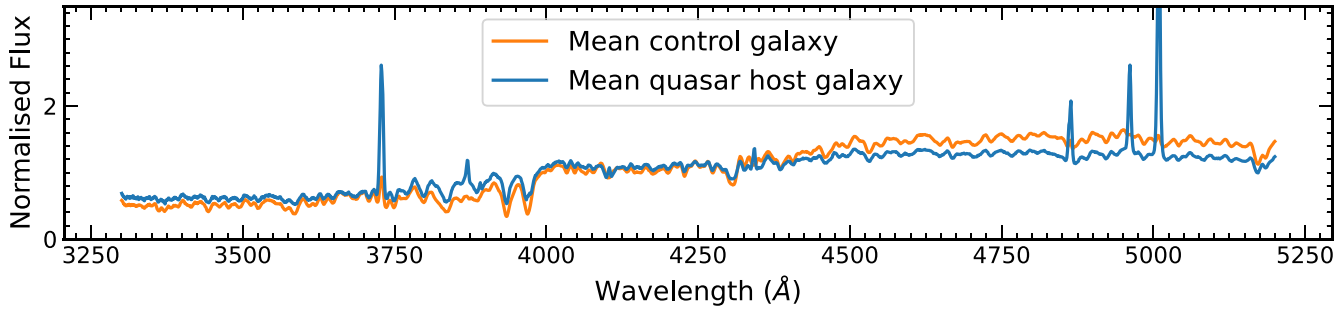


Figure 13. Comparison between the stack (mean) of all quasar host galaxy reconstructions (blue) and the stack of all control galaxy reconstructions (orange). The quasar host galaxy stack has weaker 4000 Å break strength, indicative of younger stellar populations. Both stacks are normalized to mean flux of unity. Note that due to the way our MFICA components are constructed, the narrow emission lines in the quasar host spectrum may come from a range of sources that are not only H II regions.

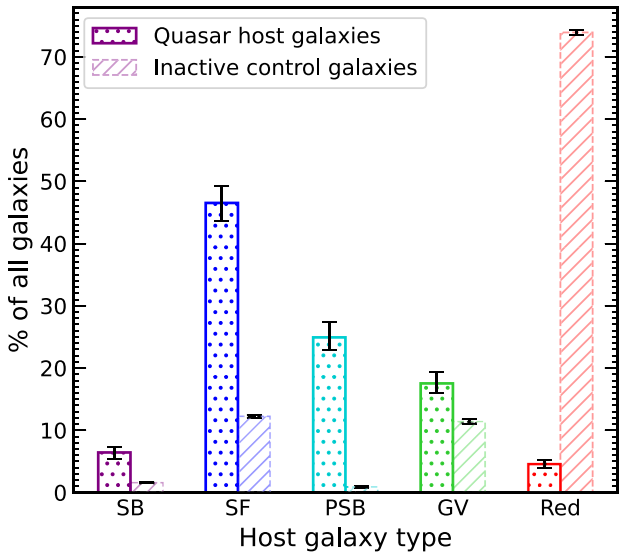


Figure 14. Percentage of different types of galaxies in the decomposed quasar host galaxy sample (dot-filled bars) and the control sample (line-filled bars). The height of each bar is the per cent of all quasar host galaxies or control galaxies with a given galaxy type. The error bars represent the maximum and minimum possible per cent of objects of a given galaxy type, based on 10 000 re-samples of the component fractions using the covariance matrix (see Section 4.1).

Table 1. Statistics of quasar hosts with reliable decompositions ($f_{gal} \geq 0.2$) and control sample galaxies. From left to right: (1) The galaxy type based on the AF and K-component fractions. (2) F_{quasar} is the per cent of all quasars that are of a given host galaxy type. (3) $F_{control}$ is the per cent of all control galaxies that are of a given galaxy type.

Galaxy type	F_{quasar} (per cent)	$F_{control}$ (per cent)
SB	6.4 ± 0.3	1.59 ± 0.03
SF	46.5 ± 0.7	12.22 ± 0.06
PSB	24.9 ± 0.6	0.90 ± 0.03
GV	17.5 ± 0.4	11.37 ± 0.11
Red	4.6 ± 0.2	73.92 ± 0.09

post-starburst galaxies studied here (Wild et al. 2016; Rowlands et al. 2018). For our $z < 0.3$ quasar subset, the post-starburst fraction decreases from 24.9 ± 0.6 per cent to 16.7 ± 0.9 per cent, while the fraction of post-starbursts in the control sample remains the same within errors. This result implies a substantial excess (∼18x) of post-starburst quasar (PSQ) host galaxies compared to the control sample. Future data sets should allow both mass and redshift matching of active and inactive galaxies, which will be an important next step.

The stellar masses of the quasar host galaxies are slightly sensitive to the exact prescription for removing the galaxy host contamination from the MFICA quasar templates (Section 2.4.2). This does not affect the estimated stellar populations of the quasar hosts, but due to the strong correlation between stellar mass and galaxy stellar populations, it does affect the inferred stellar populations of the inactive control sample. To investigate the impact of this on our conclusions we repeated the analysis with new quasar components which (a) did not remove the galaxy light, or (b) were created from the very brightest quasars where contamination from host light is naturally minimized. We find that the stellar masses of the quasar hosts changes by less than 0.2 dex in both cases. However, this small change alters the relative proportion of star-forming, green-valley and quiescent stellar populations between the quasars and inactive control sample. We therefore caution that, while the qualitative result holds that the quasar hosts are more star-forming than a mass-matched inactive galaxy sample, the exact amount by which they differ depends in detail on the precise stellar masses of the quasar hosts. On the other hand, the relative fraction of post-starburst galaxies between quasar hosts and inactive control galaxies does not significantly change and is therefore quantitatively robust. Further improvements on measuring the stellar masses of the quasars could come from an extended wavelength range of upcoming galaxy surveys, or combined photometric and spectroscopic decomposition techniques.

6 DISCUSSION

In this section we discuss our results in the context of previous AGN host galaxy studies and the implications for our understanding of the co-evolution of AGN and massive galaxies.

6.1 Stellar masses of quasars

Fig. 11 shows that our quasars are hosted by massive galaxies with median fibre stellar mass of $10^{10.8}\ M_{\odot}$ and aperture corrected total stellar masses of $\sim 10^{11.3}\ M_{\odot}$. It is well established that AGNs generally have massive host galaxies ($M_{*} > 10^{10} M_{\odot}$; Kauffmann et al. 2003b; Silverman et al. 2009; Trump et al. 2013). While physical explanations for this have been proposed (e.g. Wang & Kauffmann 2008; Matsuoka et al. 2015), in our sample the preference for massive

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hosts may be mostly explained by the flux-limited nature of the SDSS quasar catalogue (Schulze & Wisotzki 2011; Aird et al. 2012). Lower Eddington ratio quasars are more common than high Eddington ratio quasars (e.g. Kelly & Shen 2013; Temple et al. 2021), and low Eddington ratio objects will satisfy the $M_i < -22$ inclusion criterion only if their black hole masses are sufficiently large. Given the $M_{\rm BH} - M_*$ relation (e.g. Reines & Volonteri 2015), this corresponds to large stellar masses (Aird et al. 2012). Additionally, quasar host galaxies with lower stellar masses will have lower f_{gal}. Such objects may fall below the f_{gal} threshold for a reliable decomposition.

Independent verification of the high stellar masses of quasar host galaxies comes from clustering studies of SDSS quasars (Richardson et al. 2012; Shen et al. 2013). These studies quote a characteristic halo mass of $\sim 10^{12} - 10^{13}\,M_\odot$ that, assuming a halo mass-to-stellar mass function (e.g. Behroozi et al. 2019), implies stellar masses in the range 10^{10} to $10^{11}{\rm M}_\odot$.

Our quasar host galaxy stellar masses are, however, higher than those reported for SDSS Reverberation Mapping (SDSS-RM) quasar samples (once the difference between the 2 arcsec BOSS and 3 arcsec SDSS fibre apertures is accounted for; Matsuoka et al. 2014, 2015; Yue et al. 2018), and those reported for the SDSS quasar catalogue by Li et al. (2021). The difference compared to the SDSS-RM sample may partly be due to larger black hole masses in our sample (Shen et al. 2011), leading to higher stellar masses via the $M_{\rm BH} - M_*$ relation. Compared to other decomposition techniques, MFICA may also report higher stellar masses due to the (correct) initial subtraction of host galaxy light from the MFICA quasar components, leaving more host light available to be assigned to the galaxies. There will clearly also be significant differences depending on assumed aperture corrections. Obtaining accurate stellar masses of quasar host galaxies is clearly an important, yet challenging, problem that demands further attention. However, host stellar masses are not the main focus of this work so we leave further examination of this problem to future work.

6.2 Star-forming host galaxies of quasars

Figs 12 and 14 show that the largest fraction of quasars are hosted by averagely star-forming, but not starburst, galaxies. This result agrees with the overwhelming majority of studies that find that low-redshift ($z \lesssim 1$) optically selected quasars are hosted by star-forming galaxies (Jahnke, Kuhlbrodt & Wisotzki 2004; Jahnke et al. 2007; Rosario et al. 2013a; Trump et al. 2013; Matsuoka et al. 2014; Bettoni et al. 2015; Stanley et al. 2017; Yue et al. 2018; Li et al. 2021; Xie et al. 2021; Zhuang & Ho 2022; Ren et al. 2024). These studies use a wide variety of indicators of active star formation, including IR luminosities (e.g. Rosario et al. 2013a; Stanley et al. 2017), optical colours (e.g. Matsuoka et al. 2014), emission-line SFRs (e.g. Zhuang & Ho 2022), and D_n4000-based stellar ages (Ren et al. 2024). Thus, the predominantly star-forming nature of quasar host galaxies is robustly established. The new result in this paper is the ability to construct a stellar-mass matched control sample, which demonstrates how unusual the star-forming nature of quasar host galaxies actually is, given their stellar masses. We can conclude that powerful quasars do not simply represent a random sub-sample of all galaxies. This in turn suggests that the presence of the AGN and young stellar population are directly or indirectly causally linked.

Previous studies have suggested that stellar populations of AGNs become younger as AGN luminosity increases (Kauffmann et al. 2003b; Vanden Berk et al. 2006; Silverman et al. 2009; Trump et al. 2013). This is analogous to a correlation between AGN luminosity and SFR (Mullaney et al. 2012; Aird et al. 2019), albeit the evidence for such a correlation has been mixed (Rosario et al. 2012; Stanley et al. 2015; Mountrichas et al. 2022). To investigate this further, we can compare our high luminosity quasar hosts to the host galaxies of lower-luminosity Type-2 AGNs reported in the literature. Studies of Type-2 AGNs selected using emission line ratios have found that their hosts have SFRs below the star formation main sequence (Ellison et al. 2016), and often lie in the green-valley or are quiescent (Kauffmann et al. 2003b; Salim et al. 2007; Schawinski et al. 2007; Wild et al. 2007; Leslie et al. 2016; Lacerda et al. 2020). Comparison with our results would then suggest that stellar populations become younger as AGN luminosity increases (e.g. Kauffmann et al. 2003b; Trump et al. 2013; Georgantopoulos et al. 2023). This could occur if, for example, more molecular gas leads to more intense star formation and higher AGN luminosity (Salim et al. 2007; Aird et al. 2010; Rosario et al. 2013b; Ni et al. 2023). However, the preference for Type-2 AGNs to inhabit green-valley galaxies may be a selection effect, with strongly star-forming Type-2 AGN host galaxies leading to emission line ratios that preclude them from AGN selection (Trump et al. 2015). In such a scenario, the only difference between Type-1 and Type-2 AGN is obscuration and viewing angle. We plan to investigate differences between the stellar population of Type-1 quasars and Type-2 quasar/AGN host galaxies in a forthcoming paper.

6.3 Post-starburst galaxy–quasar connection

Fig. 14 also shows that a large fraction of quasar host galaxies are post-starburst galaxies (24.9 ± 0.6 per cent) and green-valley galaxies (17.5 ± 0.4 per cent). The former is especially significant, since it implies an excess of 28 ± 1 times compared to the number of post-starburst galaxies in the control sample. Post-starburst galaxies are known to be rare in the local Universe, contributing ∼1-2 per cent of the galaxy population (Wild et al. 2016; Rowlands et al. 2018). Their frequency rises with redshift, but only to 2–3 per cent at $z \sim 0.7$ using similar spectroscopic selection methods (Rowlands et al. 2018). Thus, even allowing for the different redshift range of our control sample, post-starburst host galaxies are significantly more common amongst our quasars than inactive galaxies.

An increased preference for post-starburst galaxies, or bursty SFHs, has been suggested by previous spectroscopic studies of both quasar host galaxies (Vanden Berk et al. 2006; Shi et al. 2009; Canalizo & Stockton 2013; Matsuoka et al. 2015; Melnick et al. 2015; Wei et al. 2018), and lower luminosity AGN host galaxies (Kauffmann et al. 2003b; Goto 2006; Georgakakis et al. 2008; Kocevski et al. 2009). Some studies further suggest that the incidence of post-starburst host galaxies increases with AGN luminosity (Kauffmann et al. 2003b; Vanden Berk et al. 2006; Cales & Brotherton 2015). Our results build on this work demonstrating a very strong link between bright quasars and post-starburst host galaxies.

Several studies have investigated the inverse question of the incidence of AGN activity in post-starburst galaxies, with some studies suggesting a low-to-normal frequency (De Propris & Melnick 2014; Meusinger et al. 2017; Lanz et al. 2022; Almaini et al. 2025), while others suggest close to ubiquitous presence of (obscured) AGNs in post-starburst galaxies (Yan et al. 2006; Wild et al. 2007; Brown et al. 2009; Yesuf et al. 2014; Pawlik et al. 2018). The difference almost certainly comes down to selection differences between different post-starburst and AGN samples. Observational studies also suggest there may be a delay in AGN activity following a starburst (Davies et al. 2007; Schawinski et al. 2007; Wild et al. 2010), further solidifying the link between AGN and post-starburst (or green-valley) hosts. While it should be possible to unify these

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two inverse approaches to identifying a link between post-starburst galaxies and AGN, it is complicated by selection effects on both sides, and will require forward-modelling of AGN and starburst visibility time-scales.

A class of objects known as PSQs have been studied previously, and are found to have black hole masses and accretion rates indistinguishable from conventional quasars (Cales et al. 2013; Cales & Brotherton 2015), with hosts showing signatures of past mergers (Brotherton et al. 1999; Canalizo & Stockton 2001; Cales et al. 2011). However, these PSQ samples are very different from our PSQ hosts, as they are typically identified via Balmer (specifically Hδ) absorption in their spectra alone (Cales et al. 2013; Cales & Brotherton 2015; Melnick et al. 2015). Because A and F stars are present in star-forming galaxies, the selection of PSQs based solely on a cut on Hδ absorption line strength does not exclude galaxies with ongoing star formation, i.e. galaxies which are not 'post'-burst (e.g. Goto 2006; Wei et al. 2018). Our study provides a census of true post-starburst galaxies among quasars. The addition of a mass-matched control sample of galaxies, analysed using the same spectral decomposition method, ensures that our results are also robust to selection effects of the post-starburst features.

Regardless of the exact nature of the stellar populations of quasar hosts, these pioneering observational studies of PSQs supported the early merger-driven quasar-galaxy co-evolution models (Sanders et al. 1988), which were widely replicated by numerical simulations (Springel, Di Matteo & Hernquist 2005; Hopkins et al. 2008; Hopkins & Hernquist 2009) and will be discussed in the following subsection.

6.4 Implications

The observed excess of PSQ host galaxies may imply that such galaxies are particularly conducive to quasar activity. Many studies find that post-starburst galaxies are strongly associated with major gas-rich mergers (Pawlik et al. 2016, 2018; Sazonova et al. 2021; Wilkinson et al. 2022; Ellison et al. 2024), and that the post-starburst phase is more common among galaxies with merger signatures than those without (Wilkinson et al. 2022; Li et al. 2023; Ellison et al. 2024), although simulations suggest that mergers are not a prerequisite for post-starburst galaxies (Pawlik et al. 2018). There is also evidence for an increase in the incidence of mergers amongst post-starburst galaxies with stellar mass (Pawlik et al. 2018; Ellison et al. 2024). At stellar masses close to those of our quasar host galaxies, Ellison et al. (2024) find that $\sim$ 90 per cent of post-starburst galaxies show signs of recent gas-rich mergers. The increased frequency of post-starburst galaxies in quasars, along with the high frequency of mergers in massive post-starburst galaxies, points to a connection between gas-rich major mergers and quasar activity.

Observational imaging studies of the connection between AGNs and mergers, via hosts with morphological disturbances, close companions and tidal tails, have yielded mixed results. Some studies find that the most luminous AGNs and quasars are more common among merging galaxies than lower luminosity AGNs (Urrutia, Lacy & Becker 2008; Ellison et al. 2019; Marian et al. 2020). Other studies have found no connection between quasars and major mergers (Hewlett et al. 2017; Villforth et al. 2017). A large part of the discrepancy between results is due to differences in experimental design. Different techniques, ranging from visual inspection (e.g. Kocevski et al. 2015), statistical (e.g. Pawlik et al. 2016) to machine learning (e.g. Bickley et al. 2021) are sensitive to different merger features which decay rapidly in brightness after the merger, affecting their detectability (Pawlik et al. 2016). Additionally, the quality of imaging data used to identify mergers differs between each study (Wilkinson et al. 2024), and identifying faint morphological features, corresponding to mergers, from under a bright quasar point spread function (PSF; e.g. Mechtley et al. 2016; Tang et al. 2023) can be quite challenging. Some studies suggest that merger signatures may be associated only with specific populations of quasars, such as red or obscured quasars, for which studies have found correlations between AGN luminosity and mergers (Urrutia et al. 2008; Glikman et al. 2015; Kocevski et al. 2015; Dougherty et al. 2024). These specific sub-populations, rather than the majority of quasars (which we find to be star-forming), may be ideal for studying the major merger-quasar connection.

Our results suggest that SMBHs showing quasar activity have two parallel feeding modes. The bulk of quasars have star-forming host galaxies with stellar populations indicative of SFHs free of intense time-limited bursts of star formation. Such objects are likely to show disc-like morphologies (e.g. Strateva et al. 2001), and such disc galaxies are likely not to have experienced a major merger in the recent past (Martig et al. 2012). In such objects, fuel for SMBH feeding may come from galactic bars (Jogee 2006), internal gas instabilities (Bournaud et al. 2012), minor mergers (Hernquist & Mihos 1995), or gas from supernovae or stellar winds (Ciotti & Ostriker 2007). Evidence for this comes from studies finding a lack of mergers among most AGN host galaxies at low redshifts (Cisternas et al. 2011; Villforth et al. 2017), disc-like morphologies of quasar host galaxies (albeit with significant bulges; Cisternas et al. 2011; Li et al. 2021; Tang et al. 2023), and a diversity of feeding mechanisms for SMBHs in simulated galaxies (Steinborn et al. 2018; Choi et al. 2024).

On the other hand, major mergers might play the leading role in triggering quasar activity in post-starburst host galaxies. Draper & Ballantyne (2012) and Hirschmann et al. (2012) showed that models with parallel SMBH feeding modes provide a good explanation for the form of AGN luminosity functions. While quasars in star-forming host galaxies may represent the bulk of SMBH growth, PSQs may represent short bursts of significant SMBH growth – an extension of the results of Wild et al. (2007). A prediction of this quasar feeding dichotomy is a higher AGN luminosity of merger-driven, PSQs compared to non-merger star-forming quasars (Hopkins & Hernquist 2009; Storchi-Bergmann & Schnorr-Müller 2019). We plan to test the proposed parallel modes of quasar feeding in future work.

Given the difficulty in determining the morphologies of high-luminosity AGN host galaxies (e.g. Mechtley et al. 2016; Zhao et al. 2022; Tang et al. 2023), we can use simplifying assumptions to compile our results quantitatively into merger fractions for quasars. To calculate the fraction of quasars fed by mergers (f_m) and fed by secular processes (f_s), we first assume that all our (high-stellar mass) post-starburst and starburst quasar hosts have experienced a recent ($<$ 1 Gyr ago) major merger (Ellison et al. 2024). By contrast, we assume that none of the star-forming (i.e. non-burst/smooth/continuous SFHs) quasar host galaxies have experienced a recent major merger. Finally, we assume that quasars in quiescent host galaxies are fed by secular processes like stellar winds, owing to low gas fractions (Ni et al. 2023). We adopt three extreme cases for green-valley quasars – (a) all are fed by major mergers, (b) all are fed by secular processes, and (c) an equal split between the two. This gives us a range of f_m and f_s values, which are shown in the Table 2. We find that f_m ranges from $\sim$ 0.3 to 0.5, conversely f_s ranges from 0.5 to 0.7. These merger fractions are consistent with those presented by Marian et al. (2020) ($f_m = 0.41 \pm 0.12$) and Ellison et al. (2019) ($f_m = 0.37$) for high-luminosity optically selected AGNs. These results quantify our previous assertion, that while mergers are conducive to quasar

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Table 2. Projected merger fractions of quasar host galaxies in our sample, based on their recent SFHs. From left to right: (1) f_m is the fraction of quasar hosts that might have experienced a recent major merger. (2) f_s $(= 1 - f_m)$ is the fraction of quasars that are not expected to have experienced a recent major merger (i.e. secularly triggered). Errors are from adding standard deviations on host type fractions in quadrature.

	Merger fraction (f_m)	Secular fraction (f_s)
Assuming all GVs are secularly fed	0.31 ± 0.01	0.69 ± 0.01
Assuming an equal split	0.40 ± 0.01	0.60 ± 0.01
Assuming all GVs are merger fed	0.49 ± 0.01	0.51 ± 0.01

activity, quasars are not primarily triggered by mergers (Ellison et al. 2019; Marian et al. 2020).

The large number of star-forming quasar host galaxies implies that a single episode of quasar activity does not always correspond to a quenched host galaxy. While such a result may not be at odds with AGN feedback (see Ward et al. 2022), this contrasts studies that support AGN feedback models owing to their hosts being in the green-valley (Schawinski et al. 2007). While older studies argued that the complete exhaustion of gas by star formation alone produces post-starburst galaxies (e.g. Wild et al. 2010; Hopkins 2012), with the quasar activity ensuring sustained quenching, this may have been due to the comparatively low resolution of the simulations at that time (Zheng et al. 2020). Most modern simulations require some form of AGN feedback to produce rapidly quenched post-starburst galaxies (Rodríguez Montero et al. 2019; Zheng et al. 2020; Lotz et al. 2021). The existence of cold gas in and around many post-starburst galaxies (Zwaan et al. 2013; French et al. 2015; Rowlands et al. 2015; Ellison et al. 2025), however, challenges the need for complete exhaustion of gas supplies. It would be interesting to compare the properties of quenching and star-forming quasar hosts to see whether a direct link between mergers and quasar activity can be found. In this regards, mergers may actually play a key role in determining which galaxies experience AGN feedback (Davies, Pontzen & Crain 2022).

7 CONCLUSION

We introduced MFICA to decompose quasar spectra. We applied MFICA to samples of SDSS DR7 galaxy and quasar spectra, obtaining data-driven templates (components) that reconstruct these spectra. The galaxy components resemble stellar populations of galaxies, and their fractional contributions allow estimation of the stellar populations, and therefore recent SFHs of galaxies. Using mock spectra, we show that MFICA can recover the fraction of galaxy light embedded in quasar spectra (f_{gal}) and, under the limits of $f_{gal} \geq 0.2$ and $S/N \geq 10$, can determine the stellar populations of quasar host galaxies.

We applied MFICA decomposition to a sample of SDSS DR7 quasars in the redshift range $0.16 \leq z \leq 0.76$, with the following main science results:

(i) Quasars are hosted by massive galaxies with a median fibre stellar mass of $10^{10.8}$ $M_\odot$ and total stellar mass of $\sim 10^{11.3}$ $M_\odot$. This agrees with e.g. clustering results that find luminous AGN to be hosted by massive galaxies.

(ii) Around 53 per cent of quasar host galaxies are star-forming or starburst galaxies, compared to 14 per cent of a mass-matched control sample of galaxies. Conversely, ~ 5 per cent of quasar host galaxies are quiescent/red, compared to 74 per cent of control galaxies. This indicates that luminous AGN are not simply a random sub-sample of the galaxy population, and young-stellar populations and AGN activity are directly or indirectly causally linked.

(iii) Around 25 per cent of all quasars are hosted by post-starburst galaxies, an excess of 28 ± 1 times compared to the number of post-starburst galaxies in the control sample. This may also be viewed as a 7.3 ± 0.4 excess of post-starbursts among quasars, for the same number of star-forming galaxies. Based on the known connection between massive post-starburst galaxies and major mergers, we argue that the connection between post-starburst hosts and luminous quasars may be mediated by major mergers.

Our results fit into a scenario whereby quasars have two parallel feeding modes: (i) a post-starburst mode of rapid black hole growth driven by mergers, and (ii) a star-forming mode driven by secular mechanisms like minor mergers and bars, where black hole growth is in step with star formation owing to a common gas supply. Under simplifying assumptions, this picture leads to a merger fraction of 30–50 per cent for quasars that is consistent with morphological studies in literature.

Finally, while our results show that a single episode of quasar activity does not typically coincide with a quenching of star formation in the galaxy, the rapid truncation of star formation in quasars with post-starburst hosts warrants further investigation given the significant implications for understanding galaxy quenching more generally.

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DATA AVAILABILITY

Data used in this paper are publicly available from the SDSS DR7 website http://classic.sdss.org/dr7/. OH skyline-subtracted SDSS DR7 spectra are available from http://www.sdss.jhu.edu/skypca/spSpec. The MFICA templates are available online via the supplementary material held by the publisher.

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APPENDIX A: EFFECT OF VARYING S/N AND f_{gal} ON COMPONENT FRACTIONS

In this section, we describe how the recovery of galaxy component fraction from mock spectra varies as a function of f_{gal} and the median S/N ratio of the spectra.

To quantify the dependence of component fraction recovery on the median S/N of spectra, we create 2000 mock spectra with a star-forming host galaxy and randomly chosen quasar spectra (see Section 4.3 for more details). We assign each spectrum a random median S/N between 5 and 20, while fixing $f_{gal} = 0.2$. In Fig. A1, we plot the per cent difference in the component fractions from decomposing the mock spectra compared to their actual values against the median S/N of the spectra. The percentage differences have been calculated with respect to the dynamic ranges of the component fractions. It is clear that as the median S/N increases, the recovery of the host galaxy component fractions improves, albeit weakly.

To quantify the dependence of component fraction recovery on f_{gal}, we create 2000 mock spectra with a star-forming host galaxy and randomly chosen quasar spectra. We assign each spectrum a random f_{gal} between 0 and 1, and fix S/N $= 10$. In Fig. A2, we plot the per cent difference in component fractions with respect to the dynamic ranges of the component fractions, against f_{gal}. The recovery of the weights improves dramatically with an increase in f_{gal}. This improvement is more substantial than a change in median S/N, highlighting that the choice of f_{gal} is more important in determining the performance of MFICA decomposition.

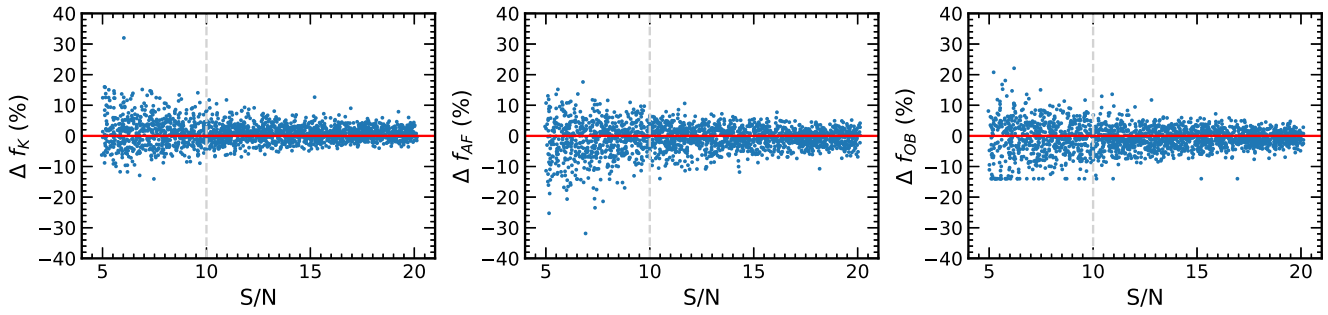


Figure A1. Scatter plots showing the per cent difference in the component fractions from decomposing mock spectra with a fixed $f_{gal} = 0.2$ and star-forming host galaxy compared to the true component fractions, as a function of the median S/N ratio of the spectra. The percentages are measured with respect to the dynamical ranges of the individual component fractions. From left to right we show the K, AF, and OB-component fractions. The dashed line represents the S/N= 10 lower limit used in our study.

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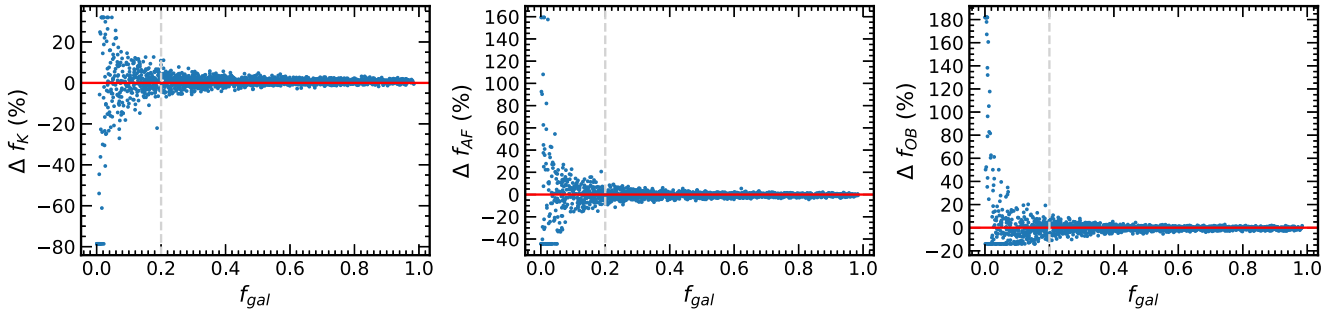


Figure A2. Scatter plots showing the per cent difference in the component fractions from decomposing mock spectra with a fixed median S/N= 10 and star-forming host galaxy compared to the true component fractions, as a function of f_{gal}. The percentages are measured with respect to the dynamical ranges of the individual component fractions. From left to right we show the K, AF, and OB-component fractions. The dashed line represents the $f_{gal} = 0.2$ lower limit used in our study.

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